

BFS Traversal

Step 1: Start at E

Legend

Complete

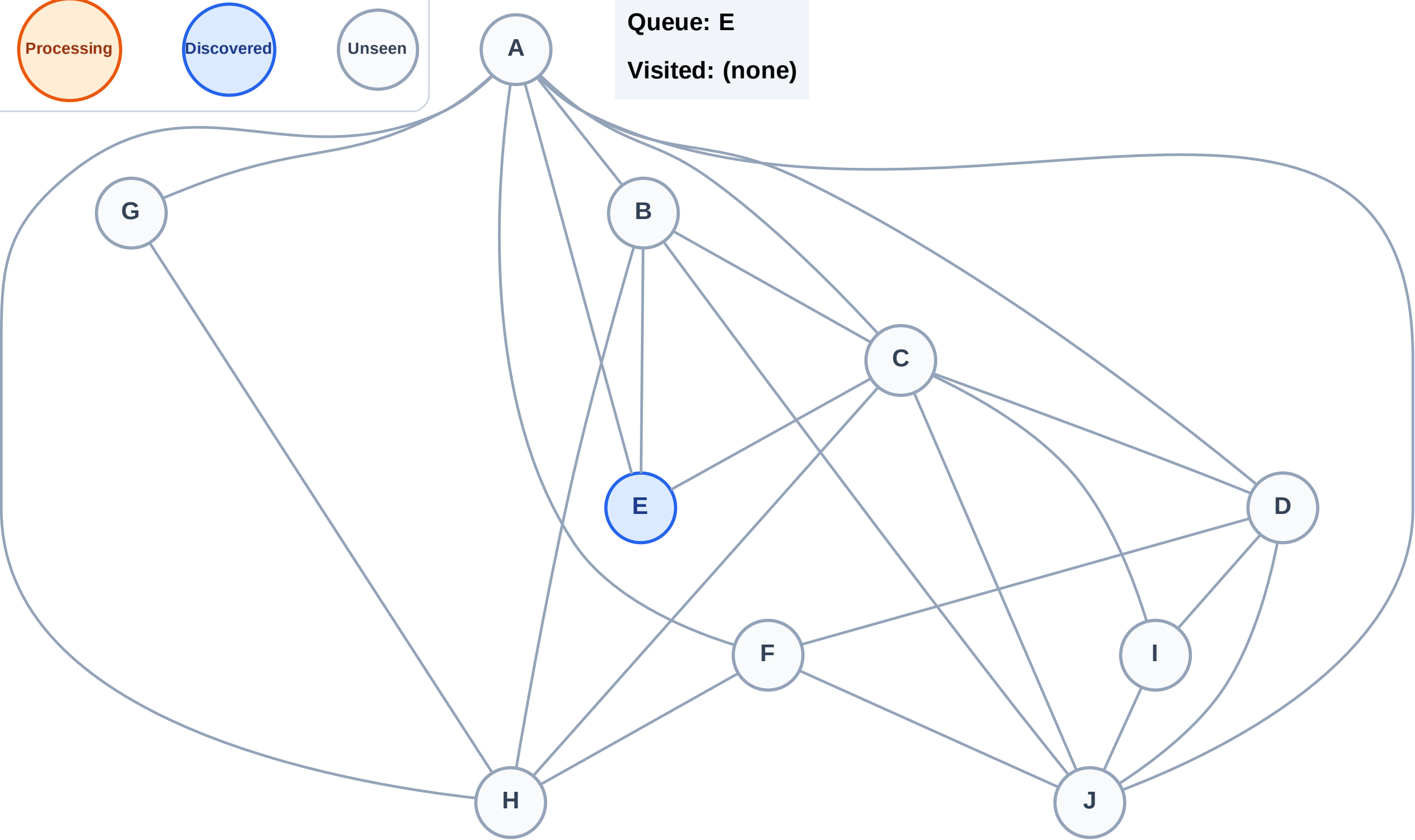
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

Complete

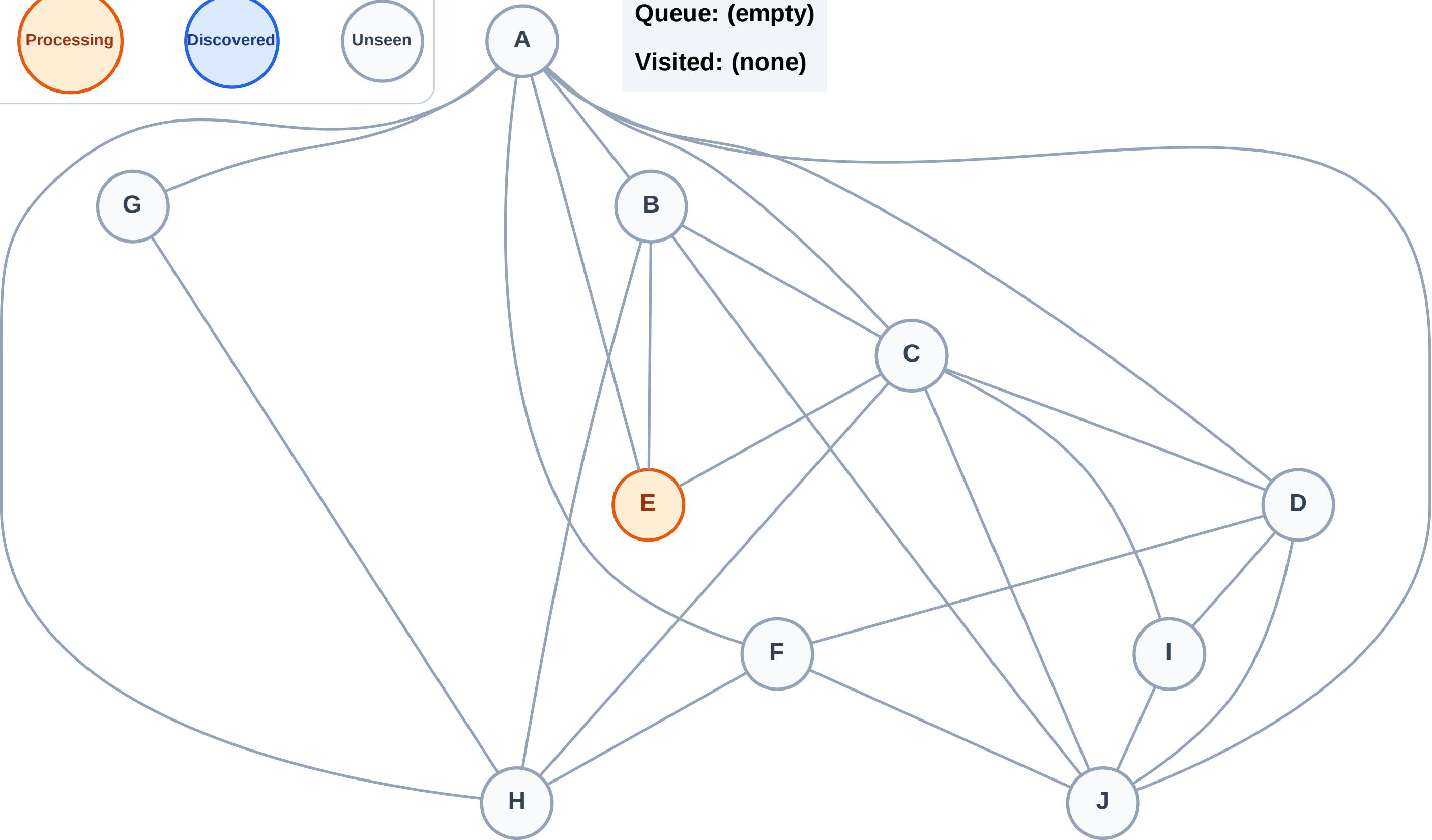
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



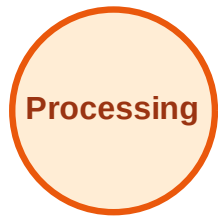
BFS Traversal

Step 3: Discover A, B, C from E

Legend



Complete



Processing



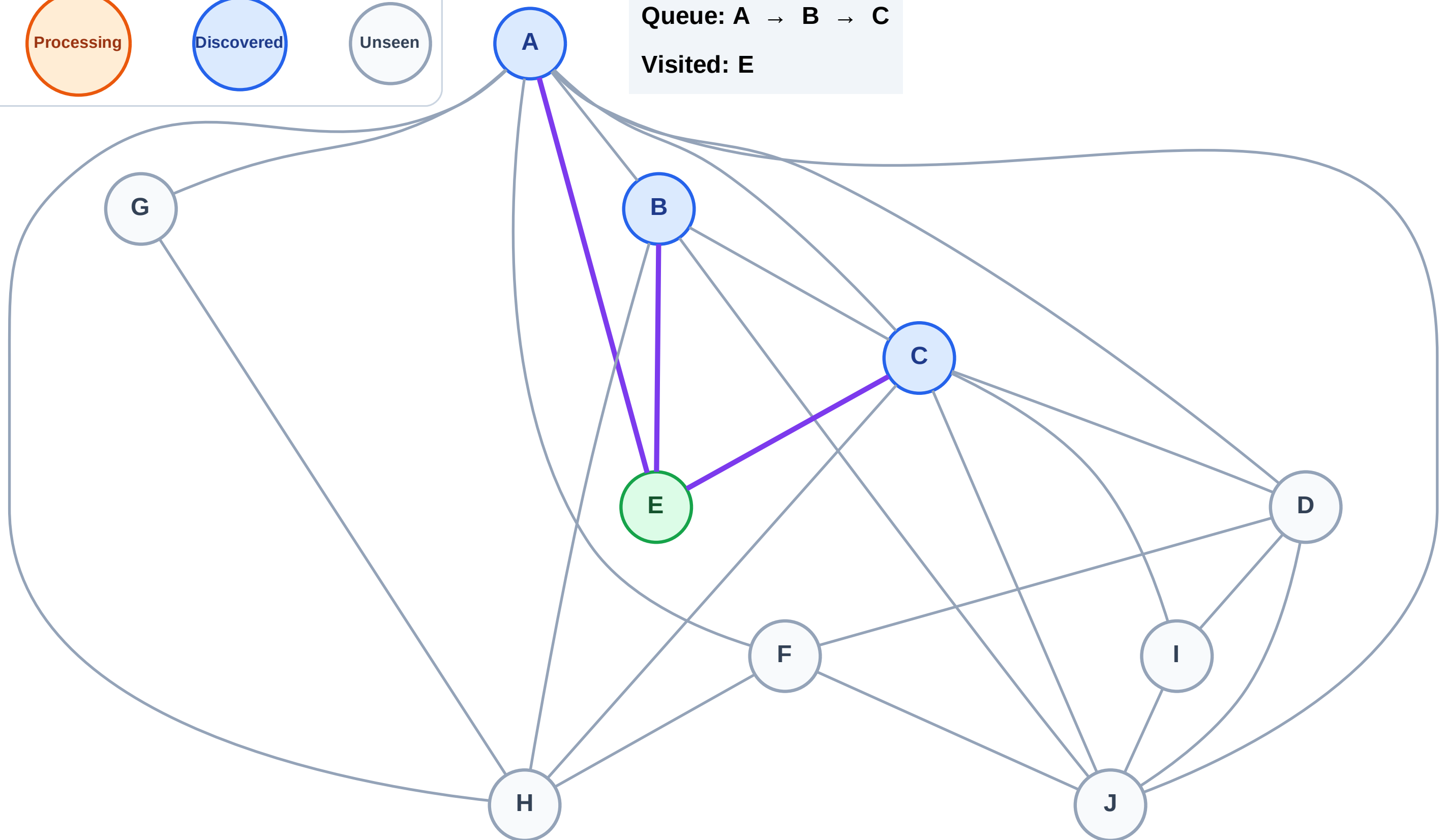
Discovered



Unseen

Queue: A → B → C

Visited: E



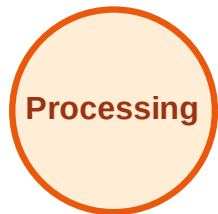
BFS Traversal

Step 4: Process node A

Legend



Complete



Processing



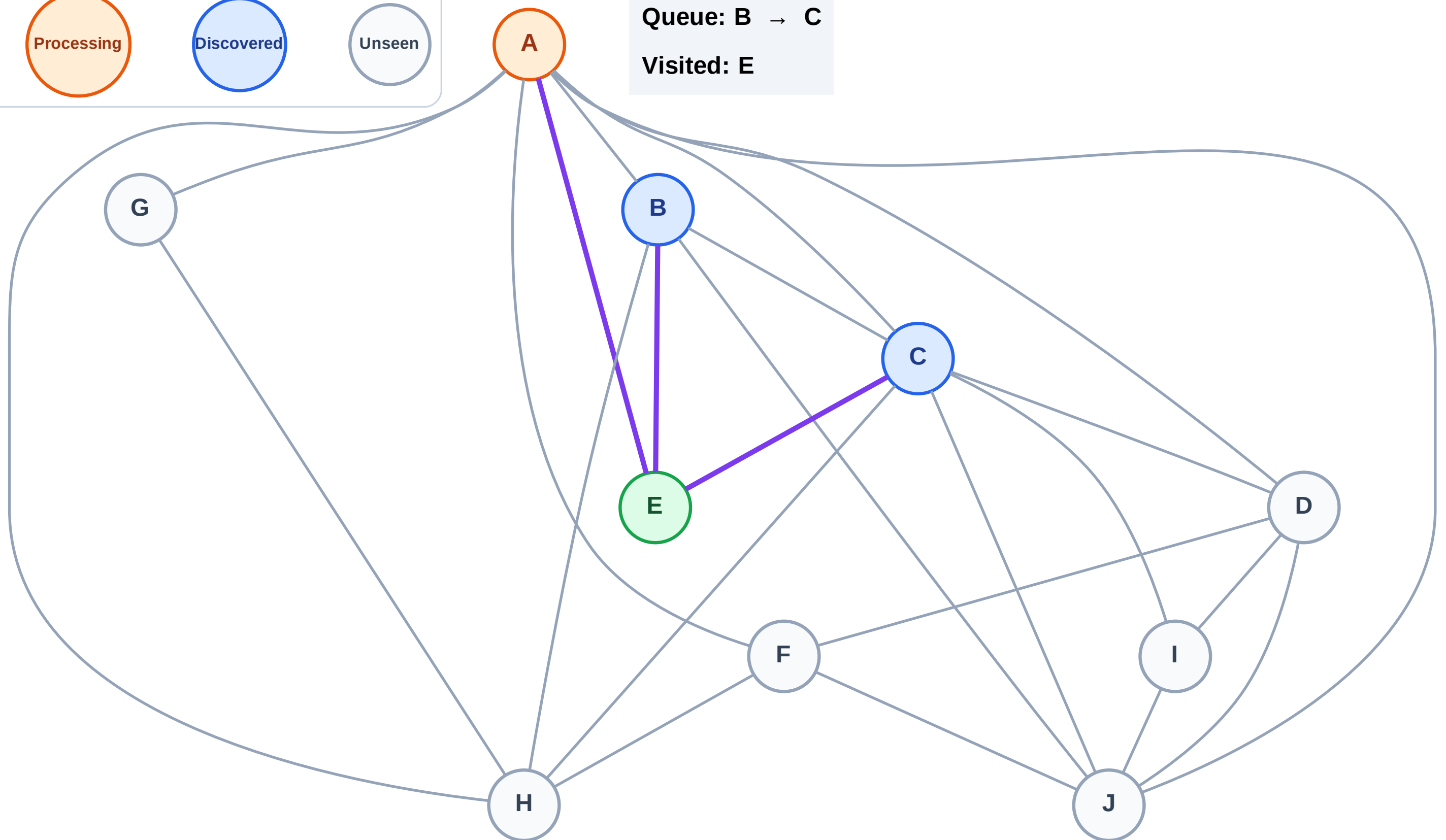
Discovered



Unseen

Queue: B → C

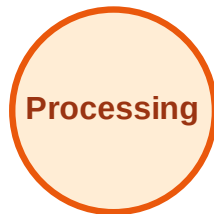
Visited: E



BFS Traversal

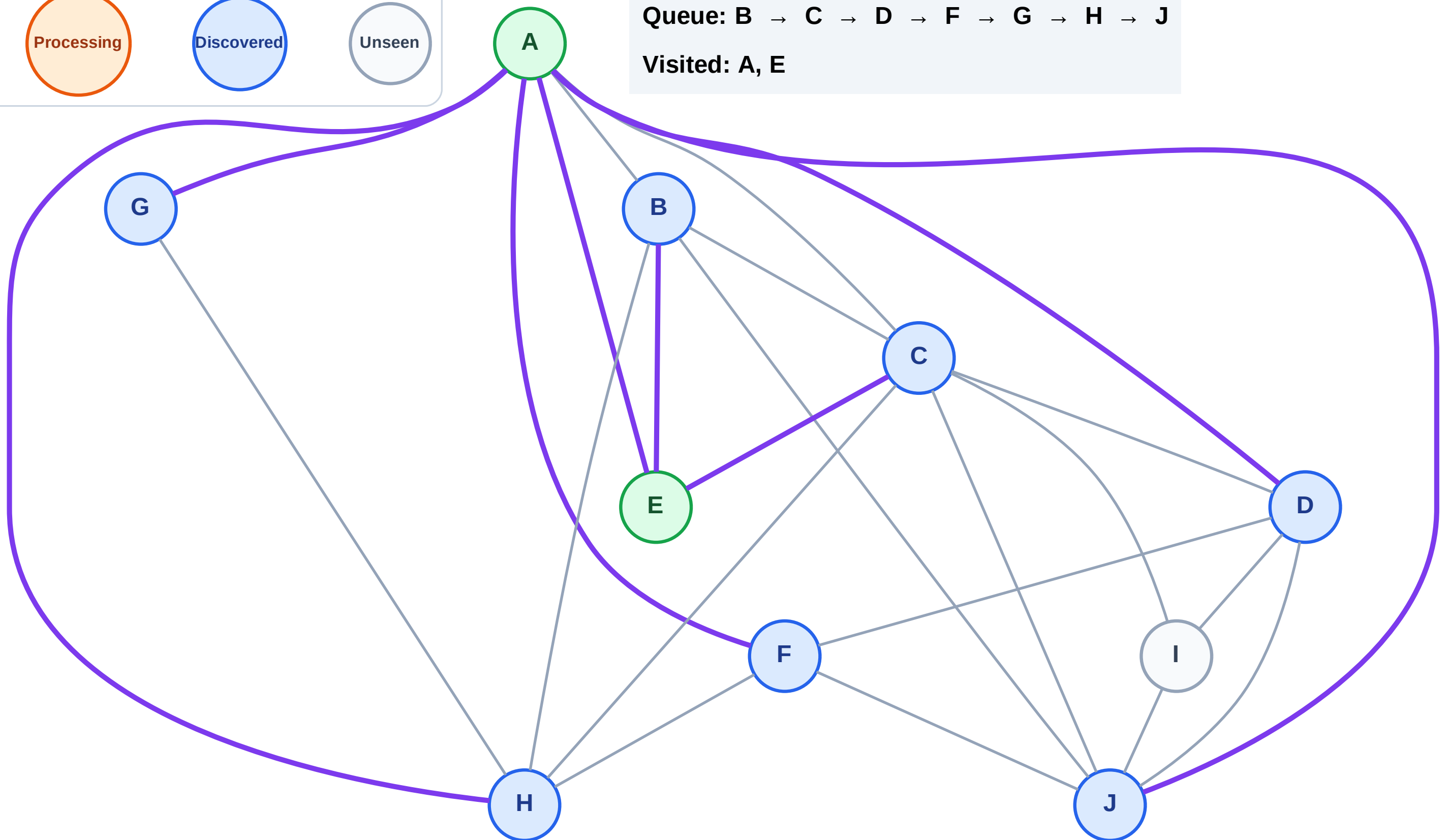
Step 5: Discover D, F, G, H, J from A

Legend



Queue: B → C → D → F → G → H → J

Visited: A, E



BFS Traversal

Step 6: Process node B

Legend

Complete

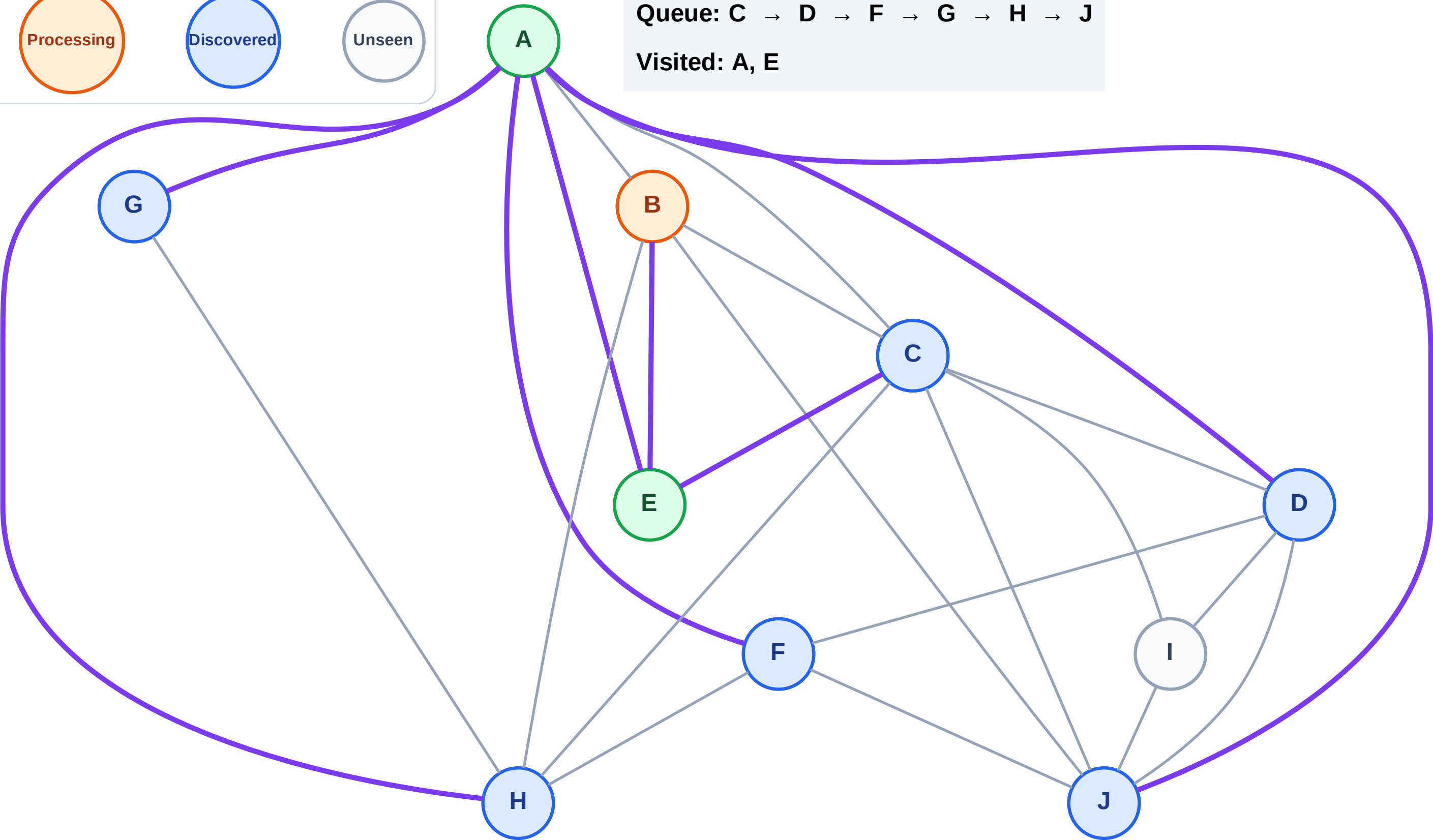
Processing

Discovered

Unseen

Queue: C → D → F → G → H → J

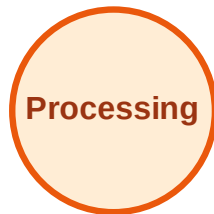
Visited: A, E



BFS Traversal

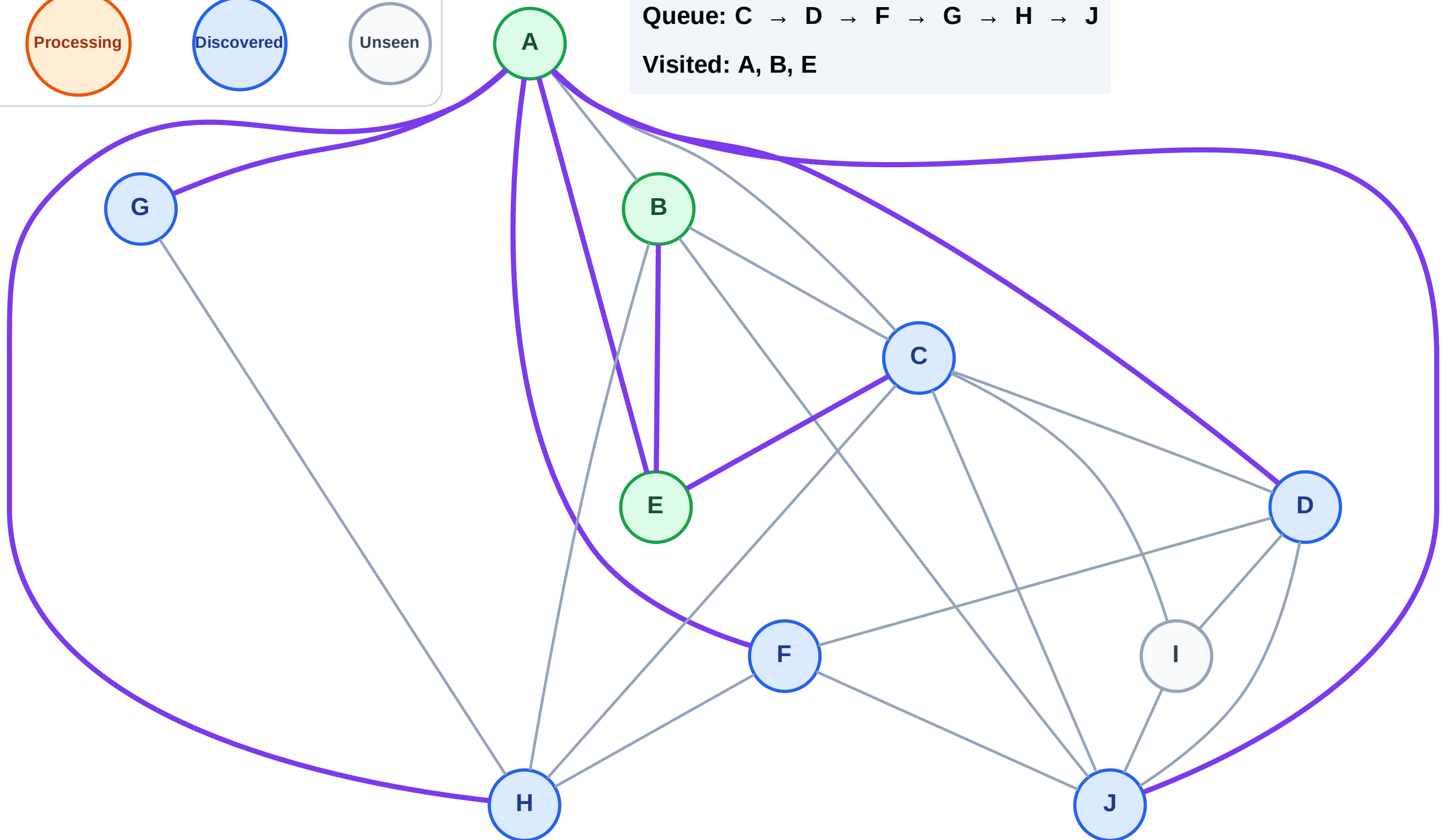
Step 7: No new neighbors from B

Legend



Queue: C → D → F → G → H → J

Visited: A, B, E



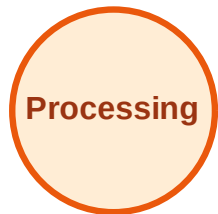
BFS Traversal

Step 8: Process node C

Legend



Complete



Processing



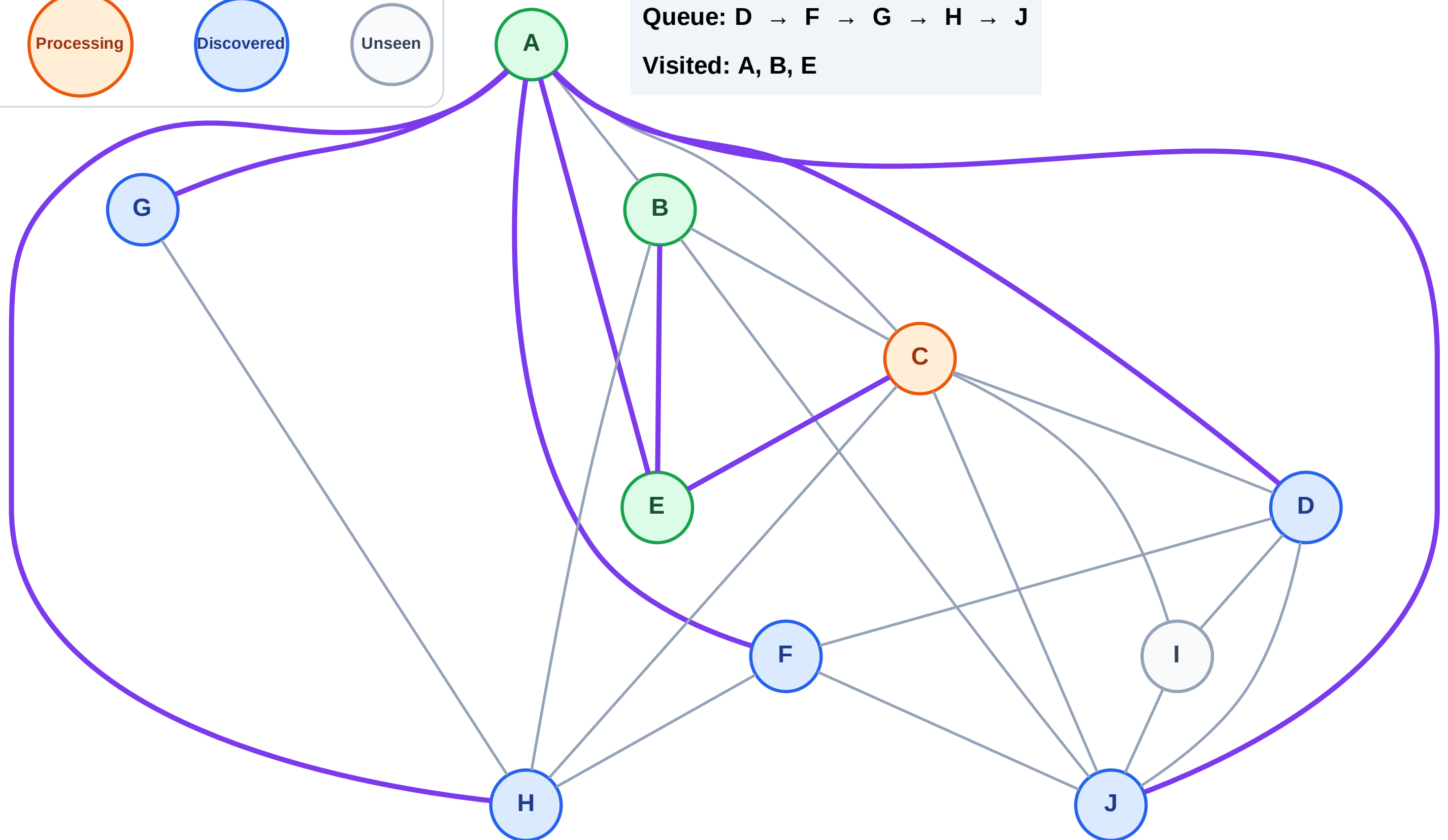
Discovered



Unseen

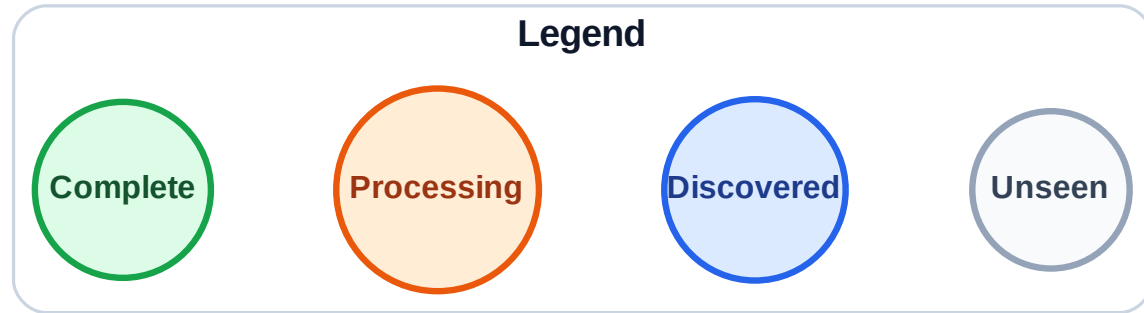
Queue: D → F → G → H → J

Visited: A, B, E

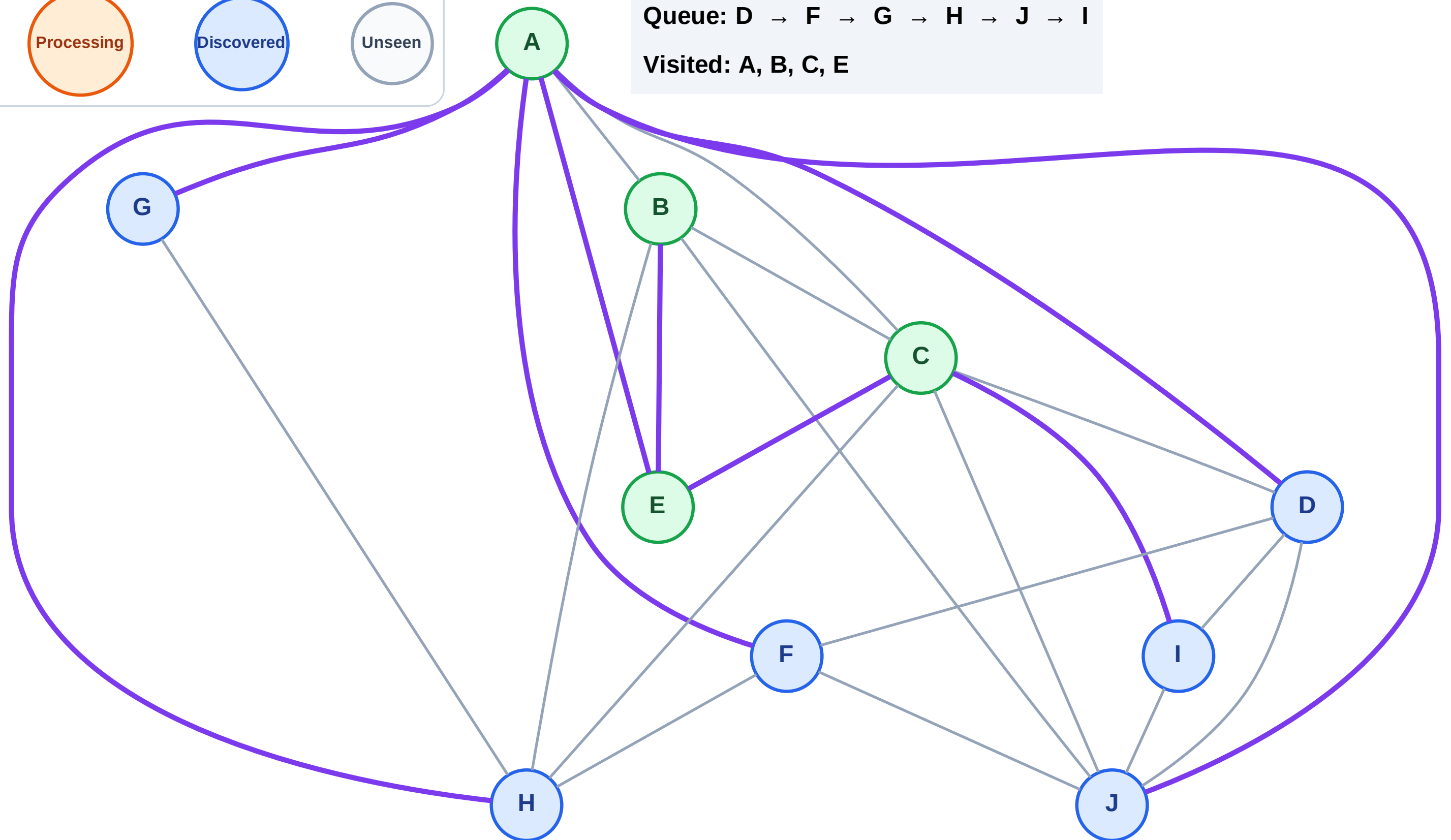


BFS Traversal

Step 9: Discover I from C



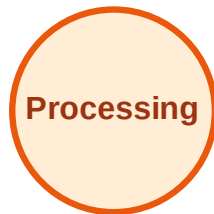
Queue: D → F → G → H → J → I
Visited: A, B, C, E



BFS Traversal

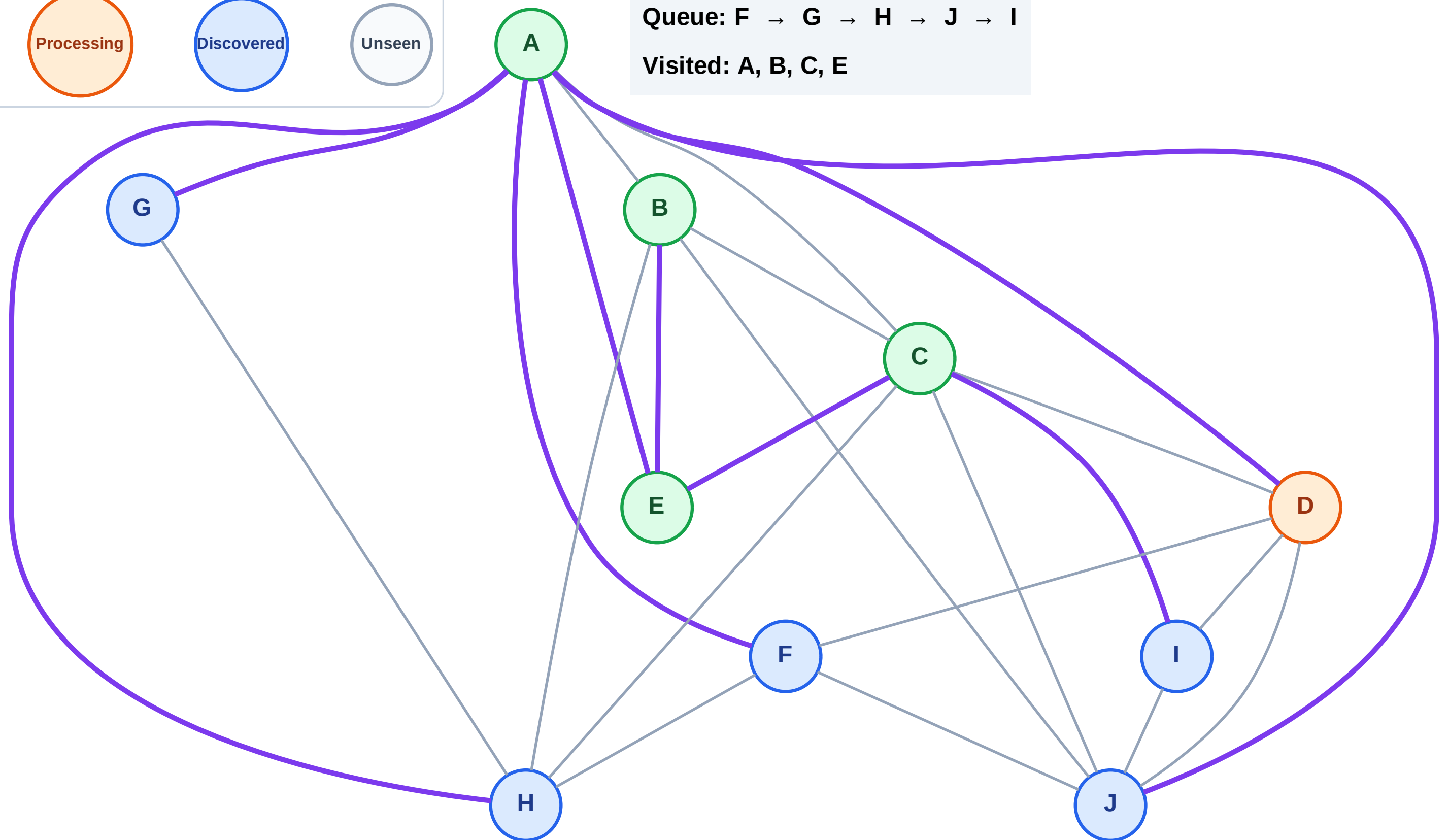
Step 10: Process node D

Legend



Queue: F → G → H → J → I

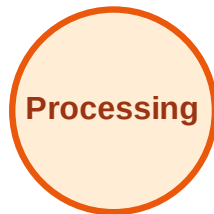
Visited: A, B, C, E



BFS Traversal

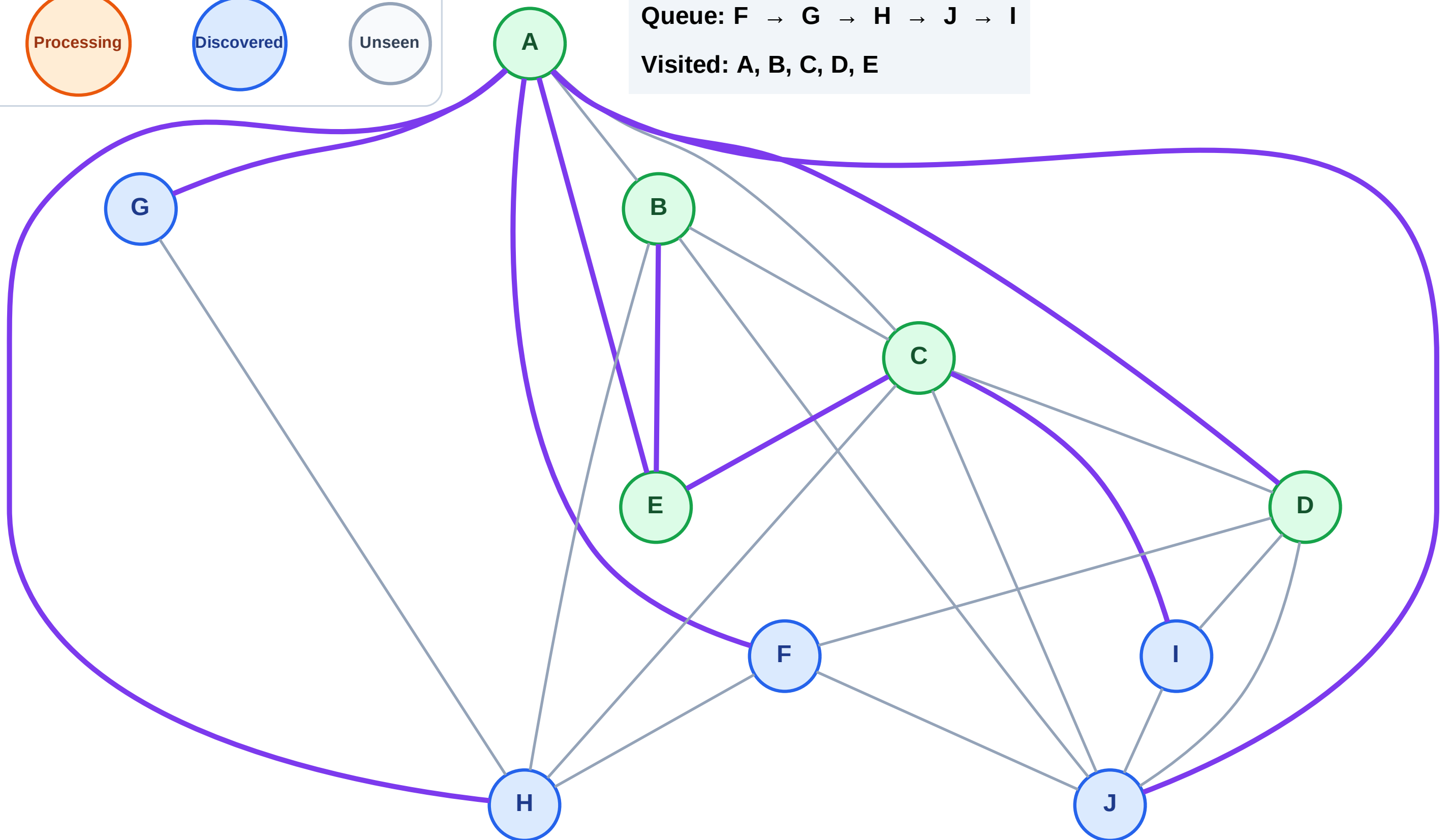
Step 11: No new neighbors from D

Legend



Queue: F → G → H → J → I

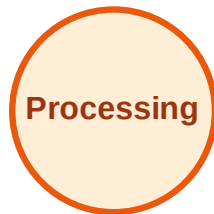
Visited: A, B, C, D, E



BFS Traversal

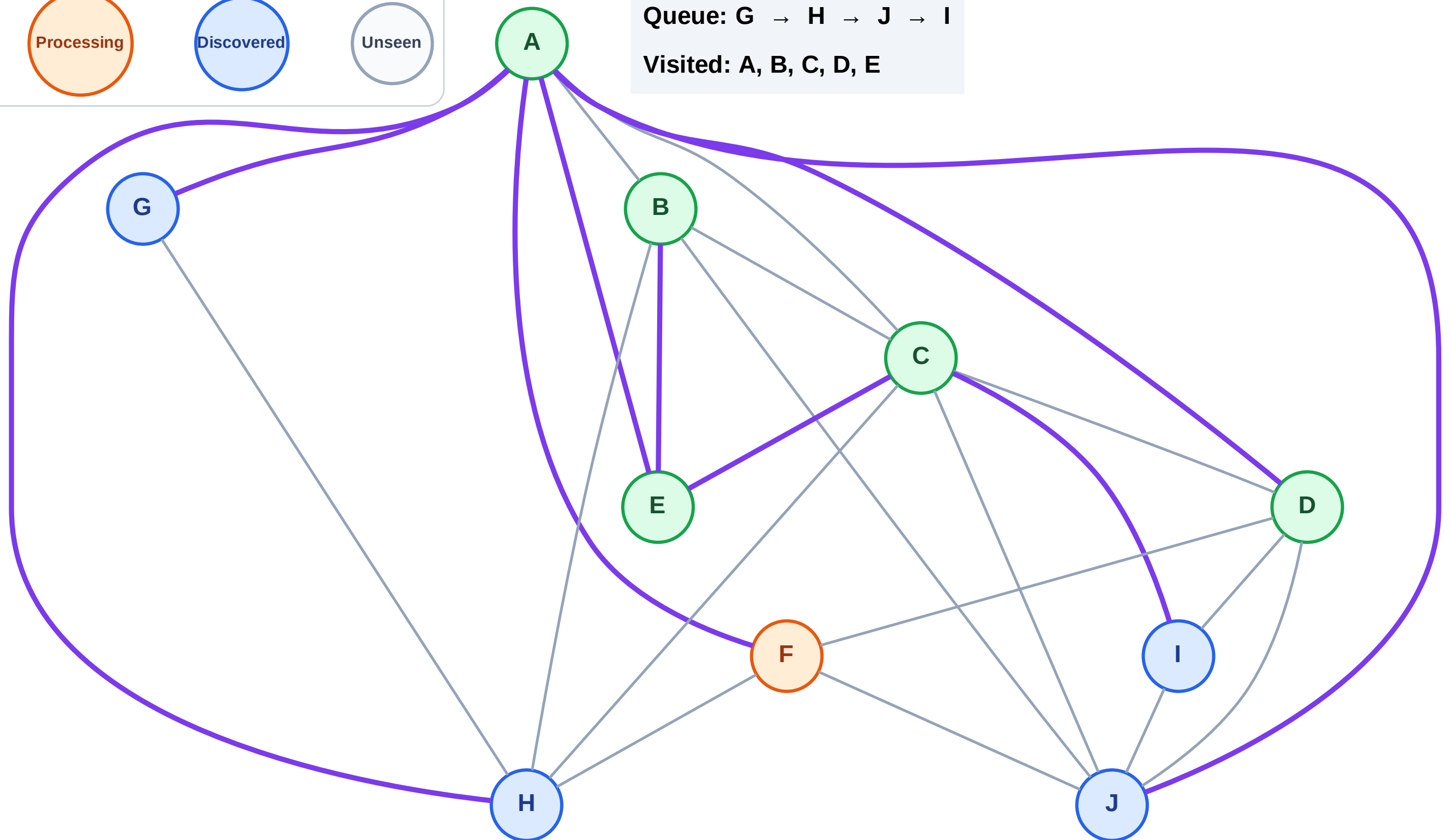
Step 12: Process node F

Legend



Queue: G → H → J → I

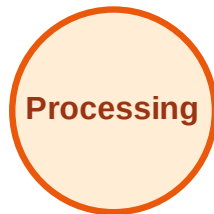
Visited: A, B, C, D, E



BFS Traversal

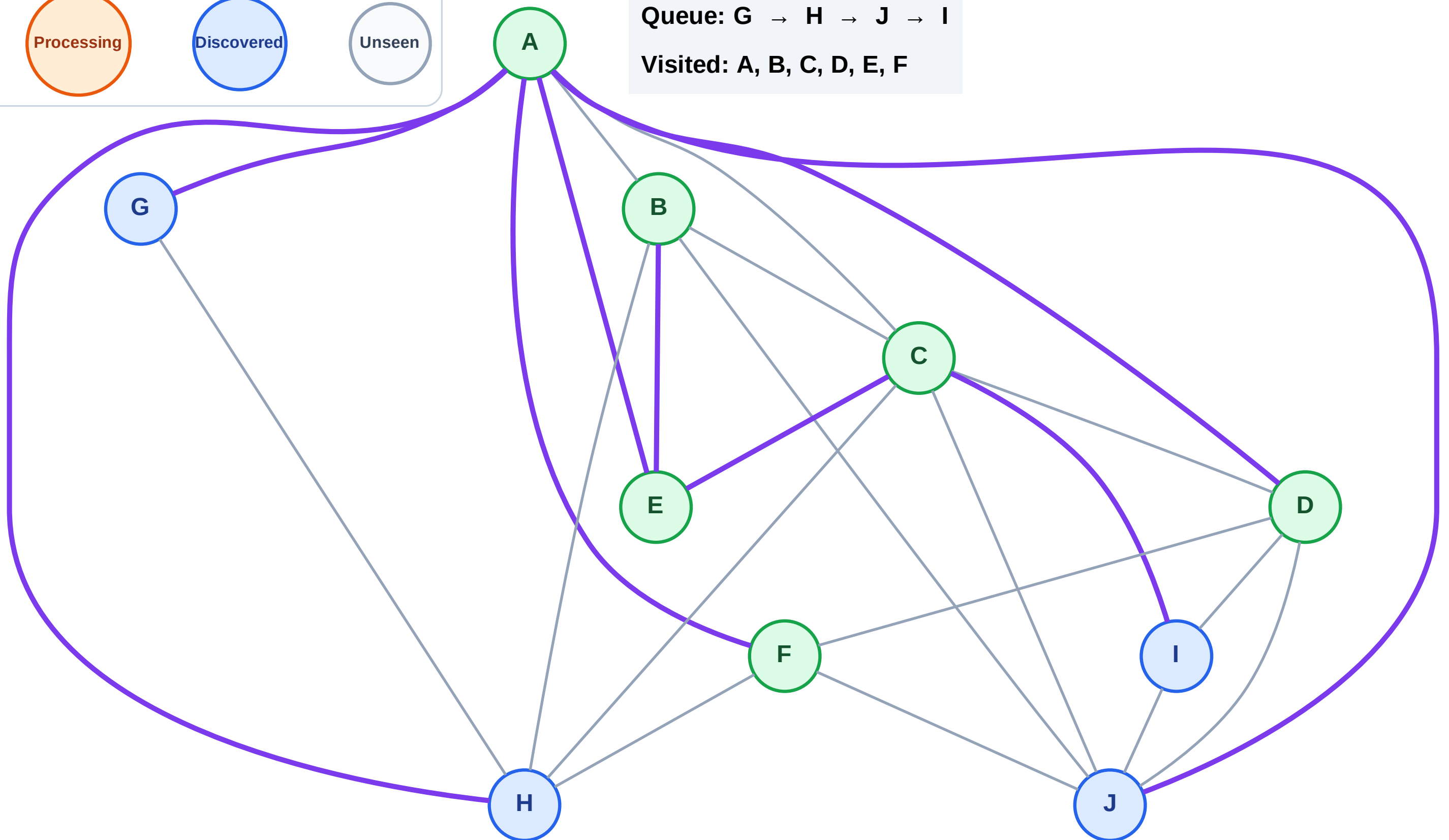
Step 13: No new neighbors from F

Legend



Queue: G → H → J → I

Visited: A, B, C, D, E, F



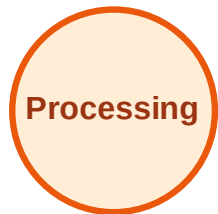
BFS Traversal

Step 14: Process node G

Legend



Complete



Processing



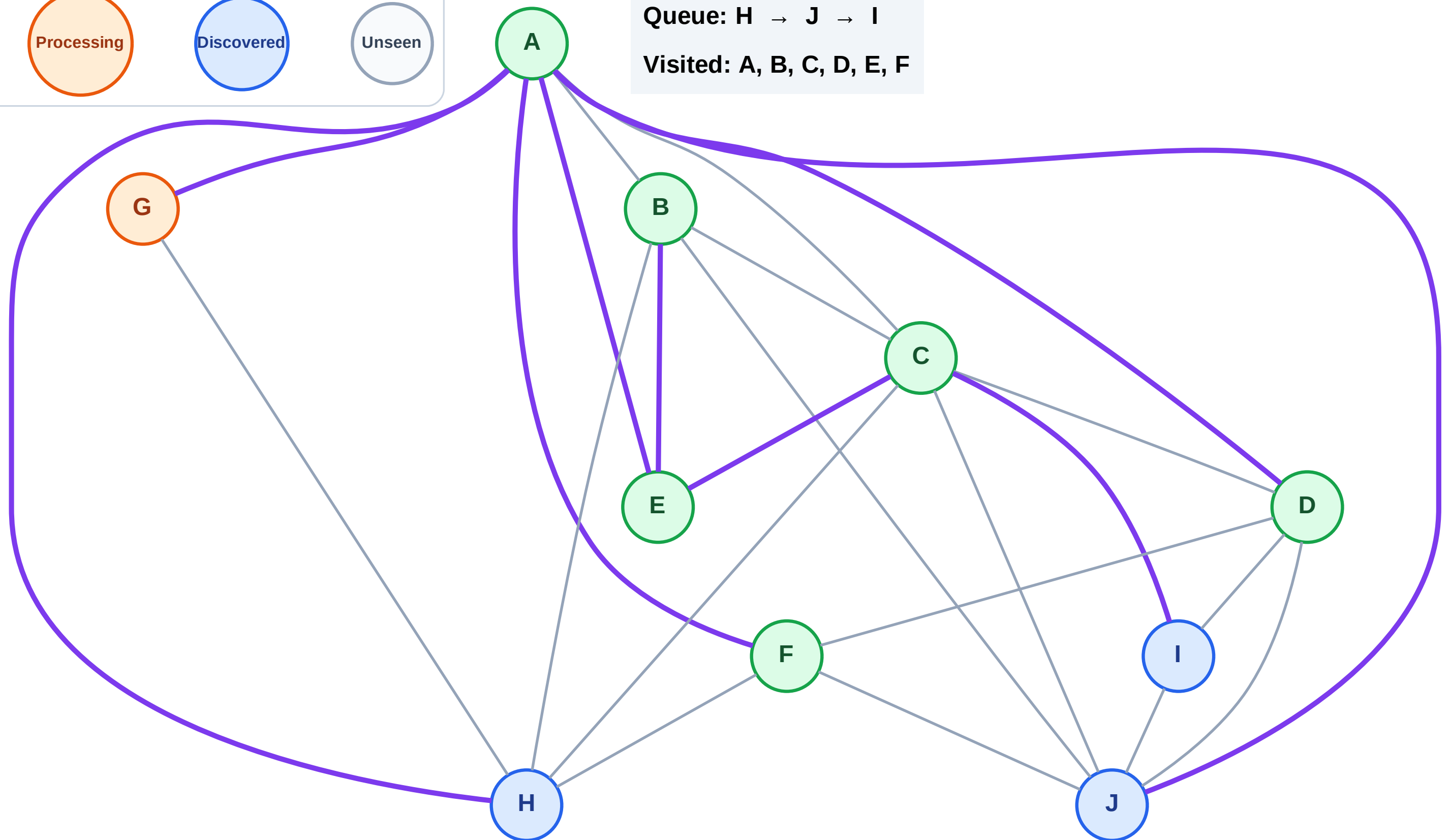
Discovered



Unseen

Queue: H → J → I

Visited: A, B, C, D, E, F



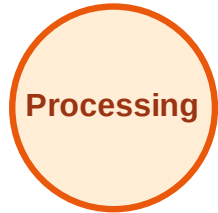
BFS Traversal

Step 15: No new neighbors from G

Legend



Complete



Processing



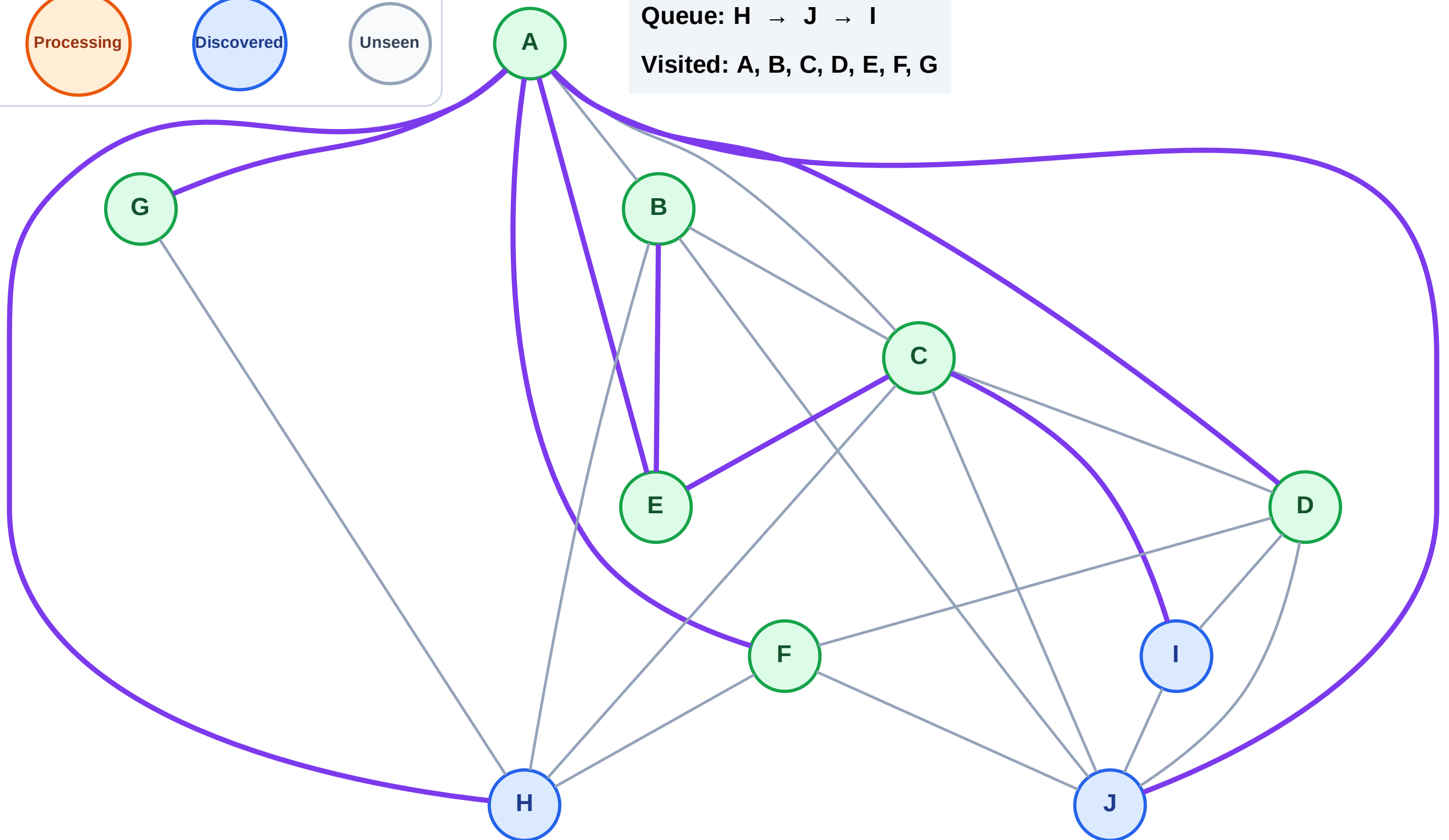
Discovered



Unseen

Queue: H → J → I

Visited: A, B, C, D, E, F, G



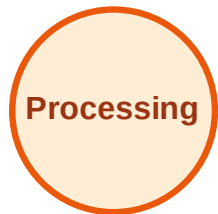
BFS Traversal

Step 16: Process node H

Legend



Complete



Processing



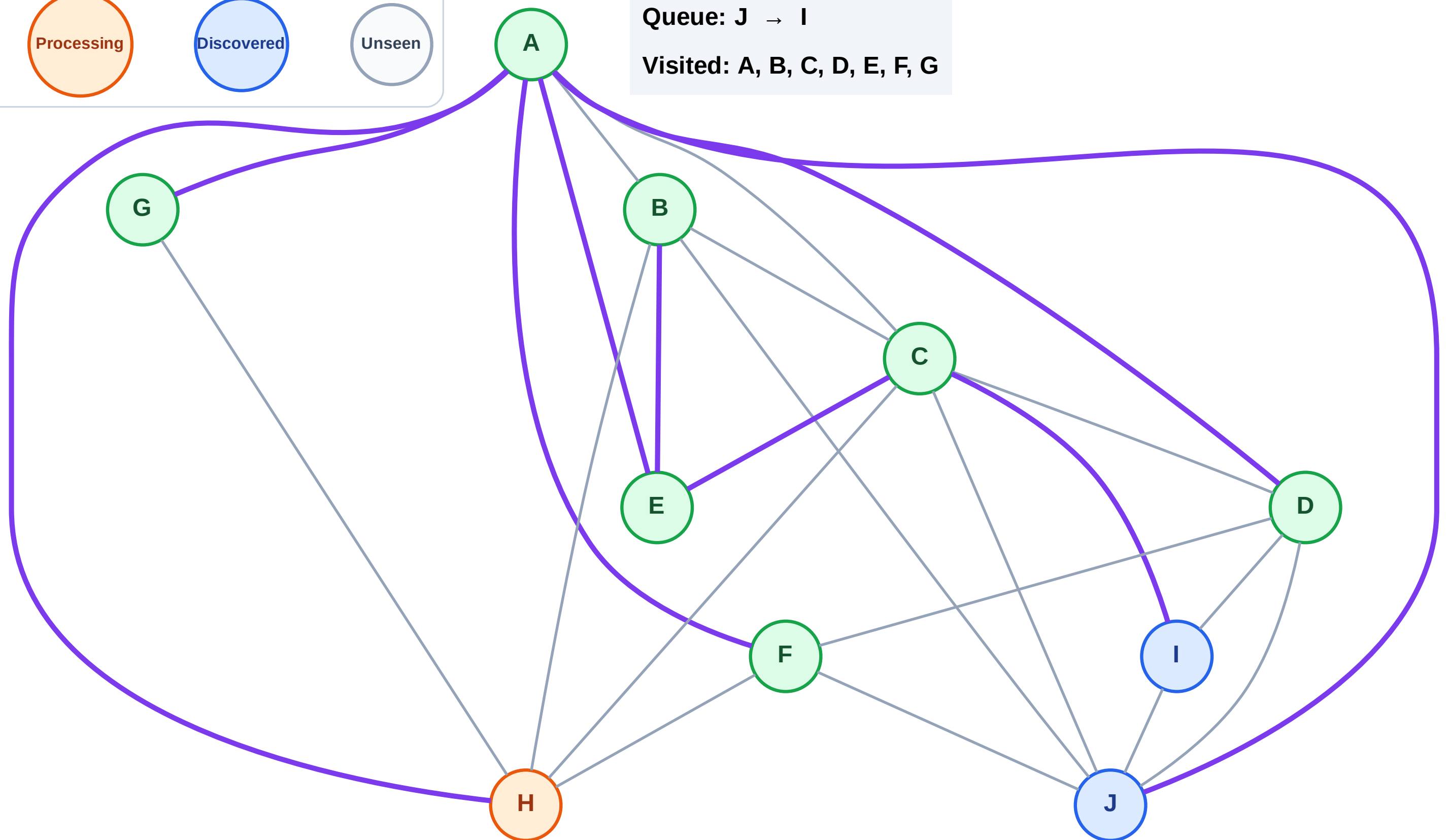
Discovered



Unseen

Queue: J → I

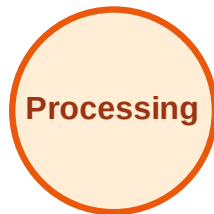
Visited: A, B, C, D, E, F, G



BFS Traversal

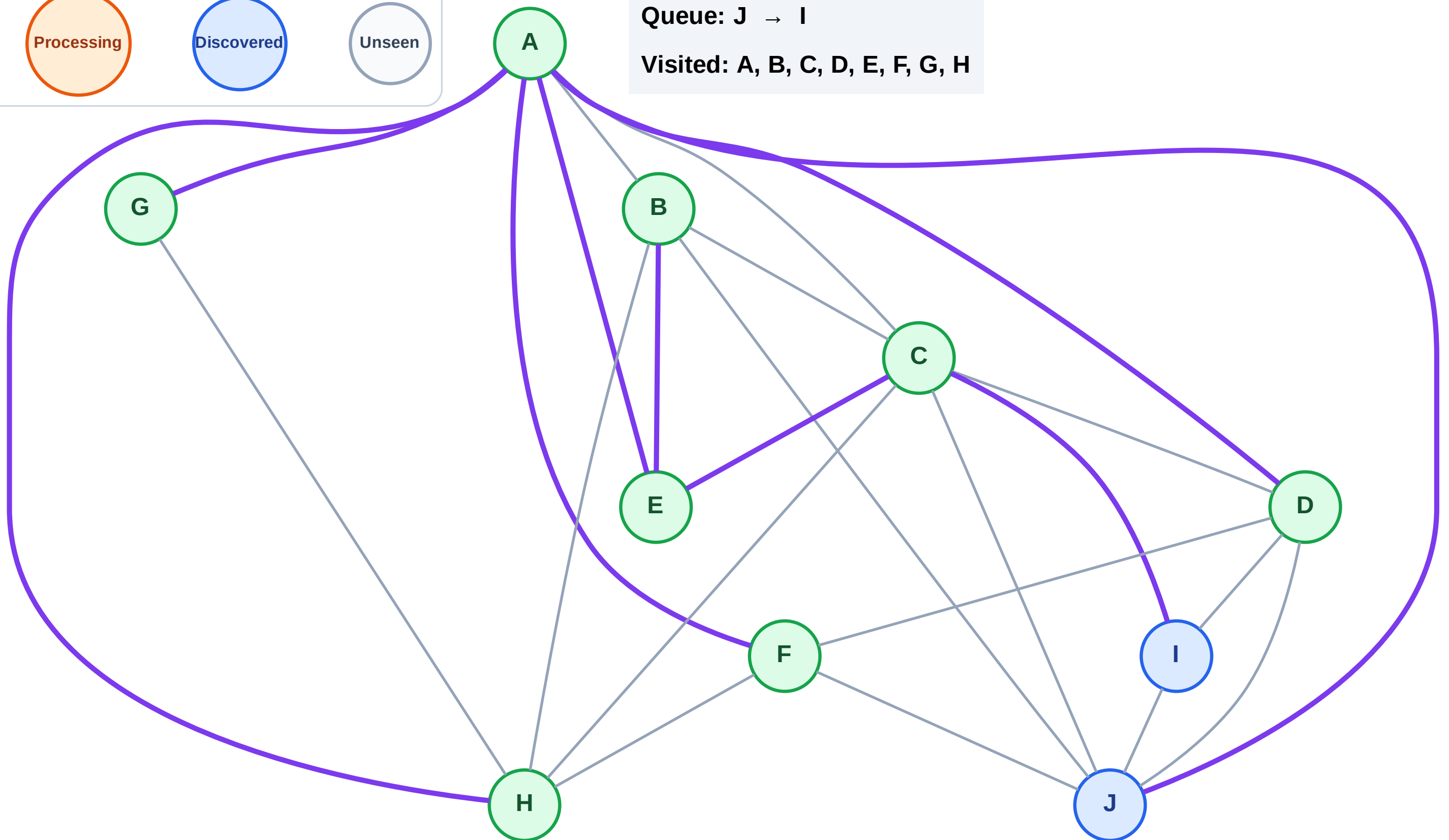
Step 17: No new neighbors from H

Legend



Queue: J → I

Visited: A, B, C, D, E, F, G, H



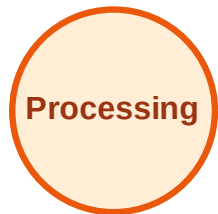
BFS Traversal

Step 18: Process node J

Legend



Complete



Processing



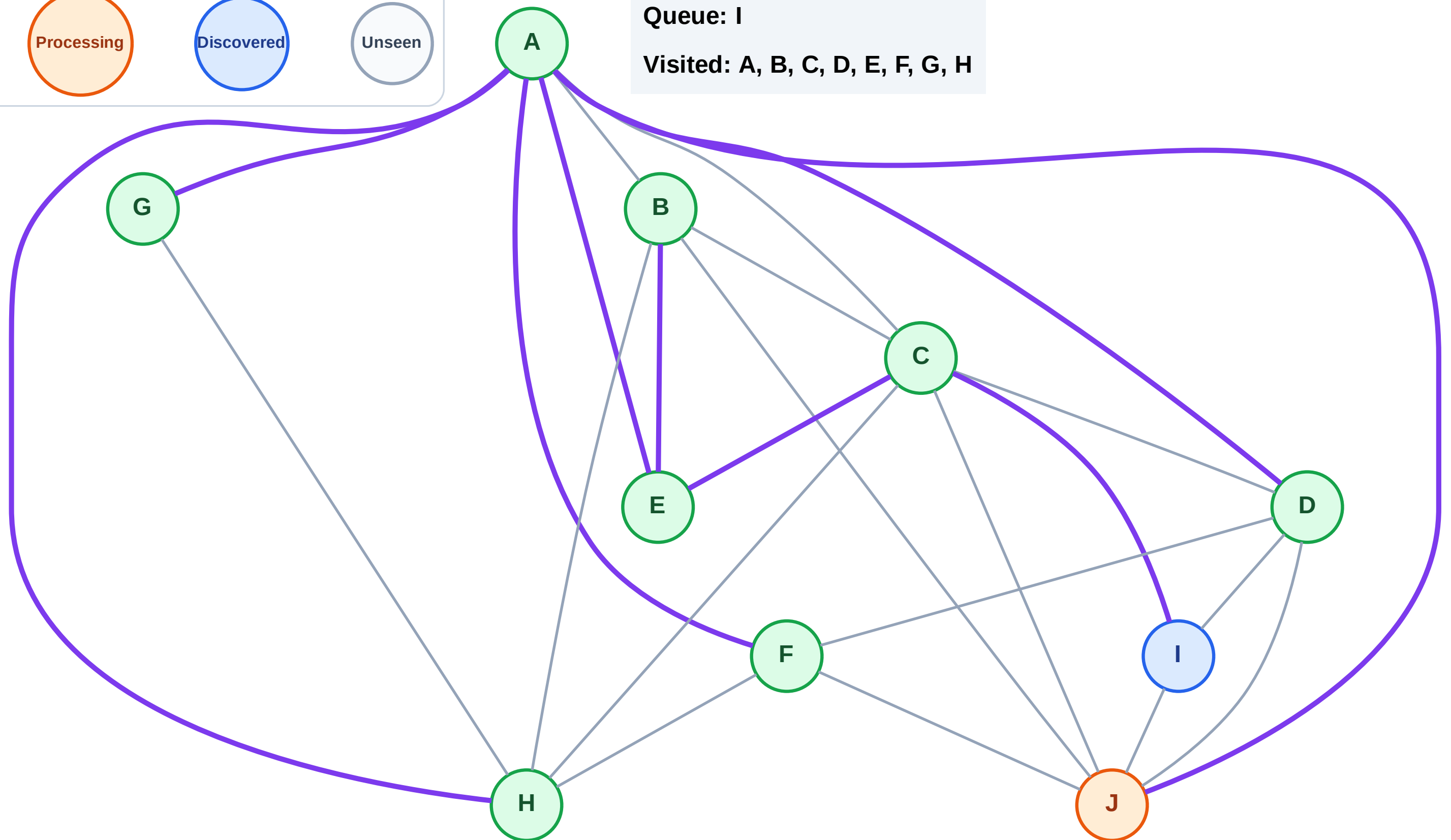
Discovered



Unseen

Queue: I

Visited: A, B, C, D, E, F, G, H



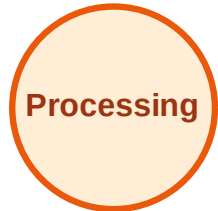
BFS Traversal

Step 19: No new neighbors from J

Legend



Complete



Processing



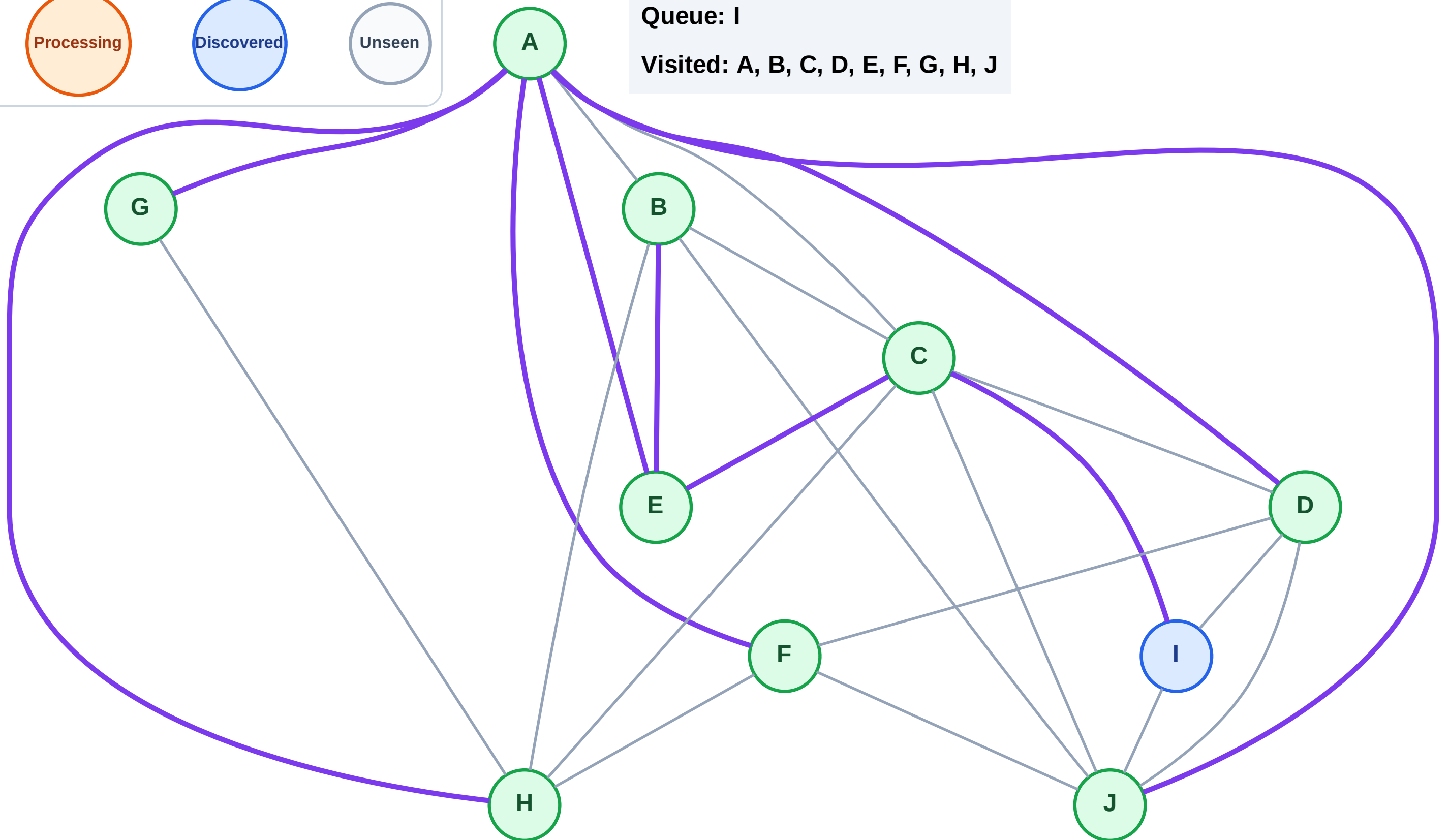
Discovered



Unseen

Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

Step 20: Process node I

Legend

Complete

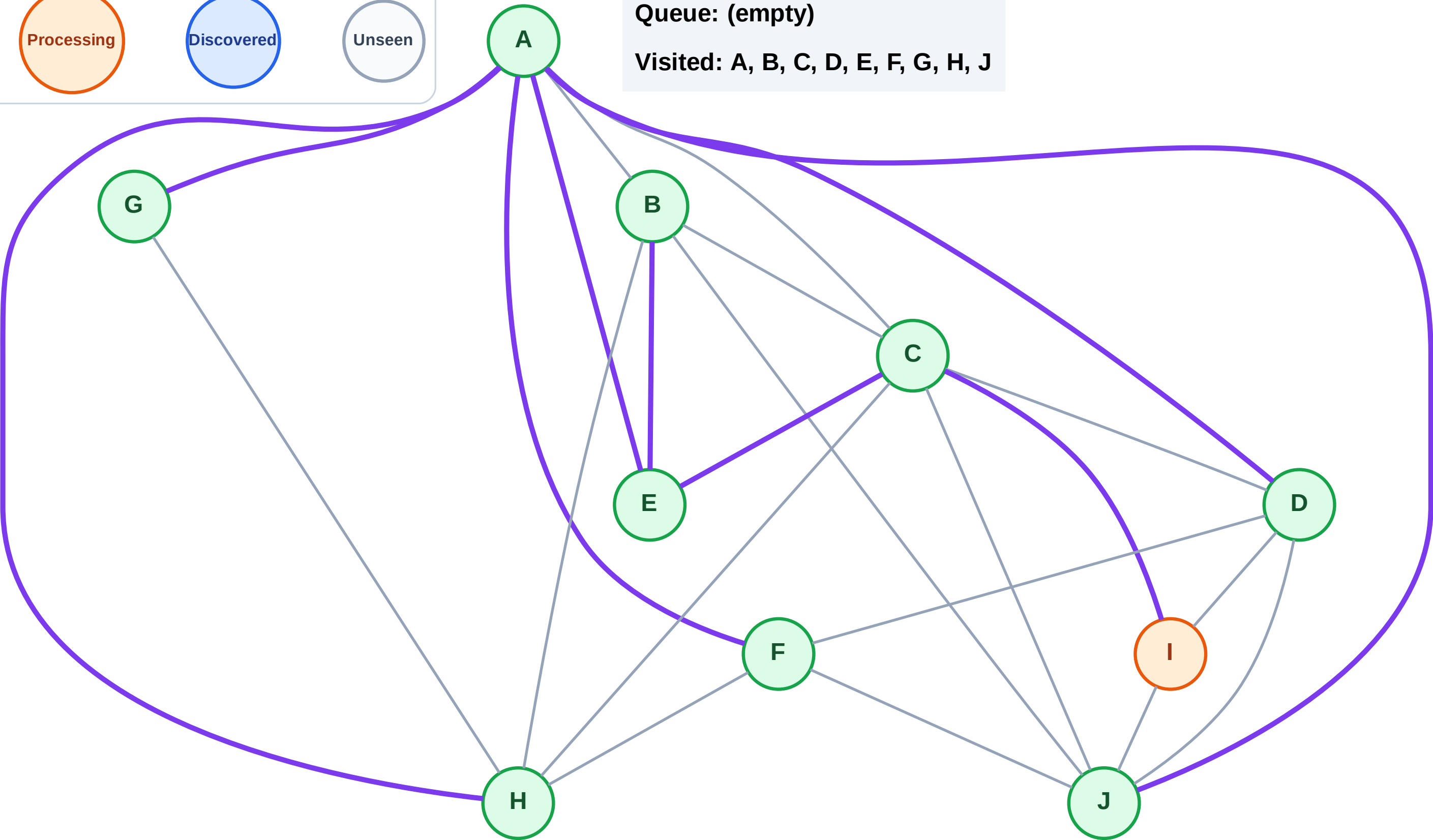
Processing

Discovered

Unseen

Queue: (empty)

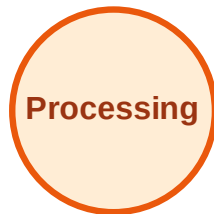
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

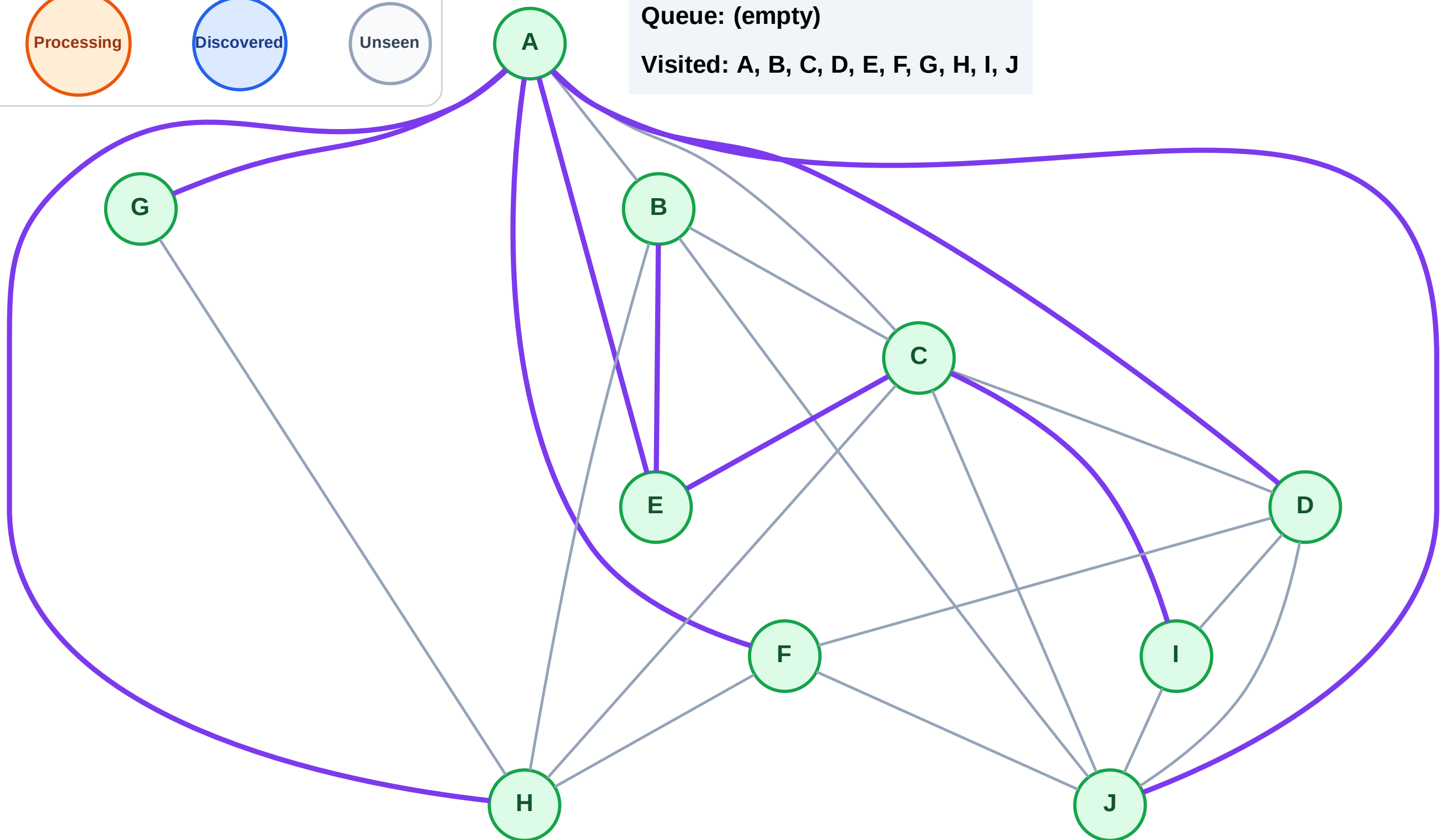
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

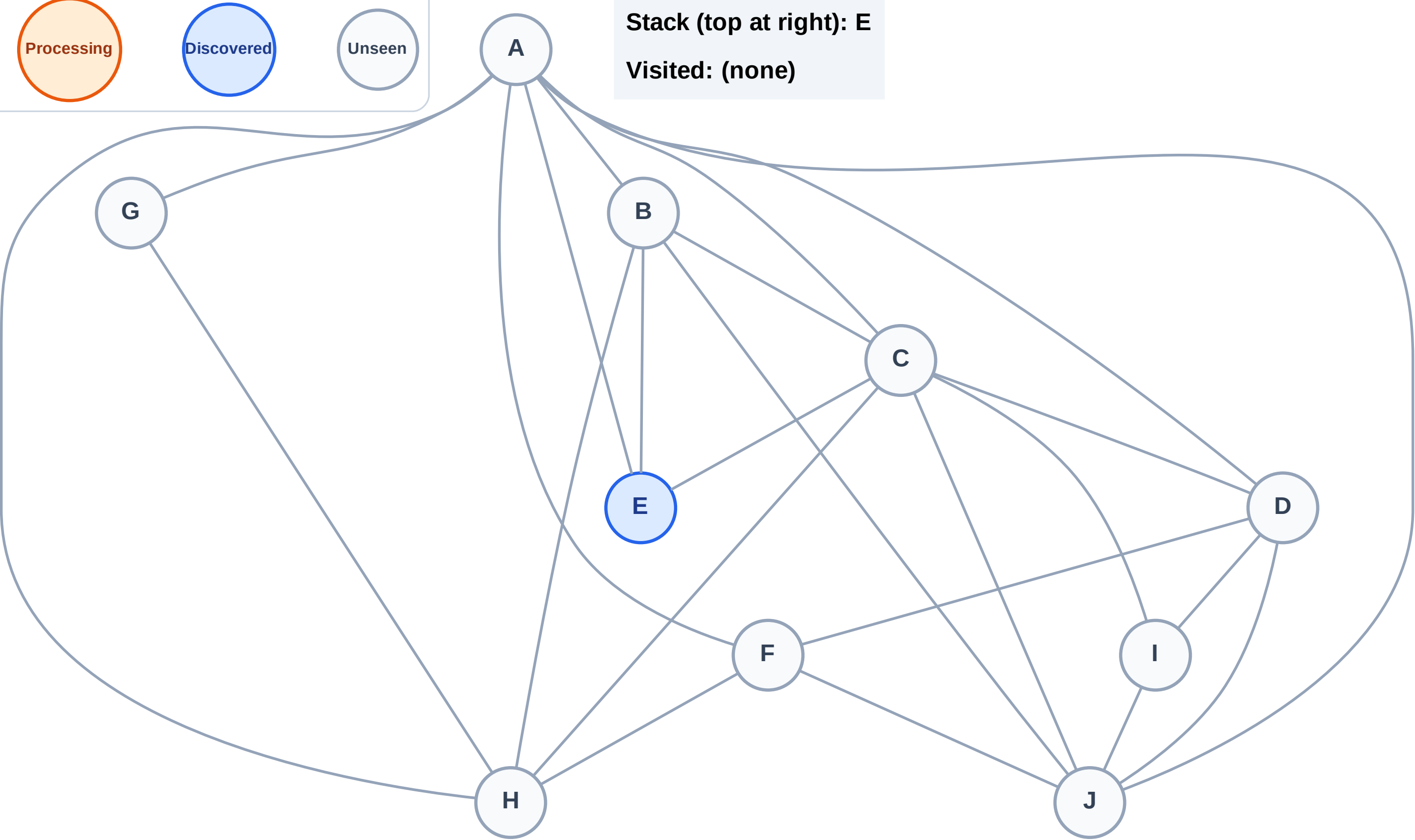
Processing

Discovered

Unseen

Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

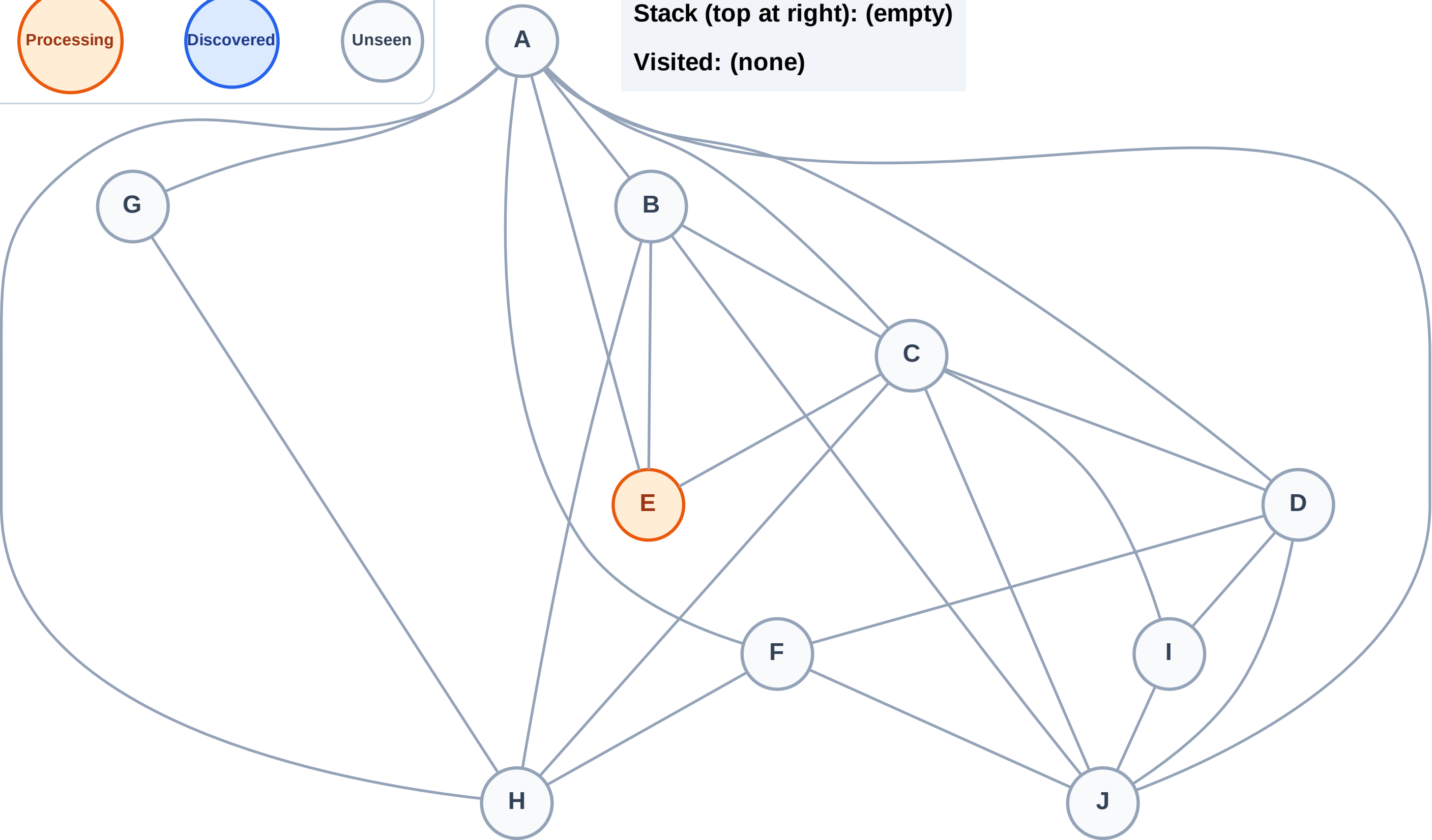
Complete

Processing

Discovered

Unseen

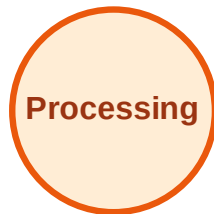
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

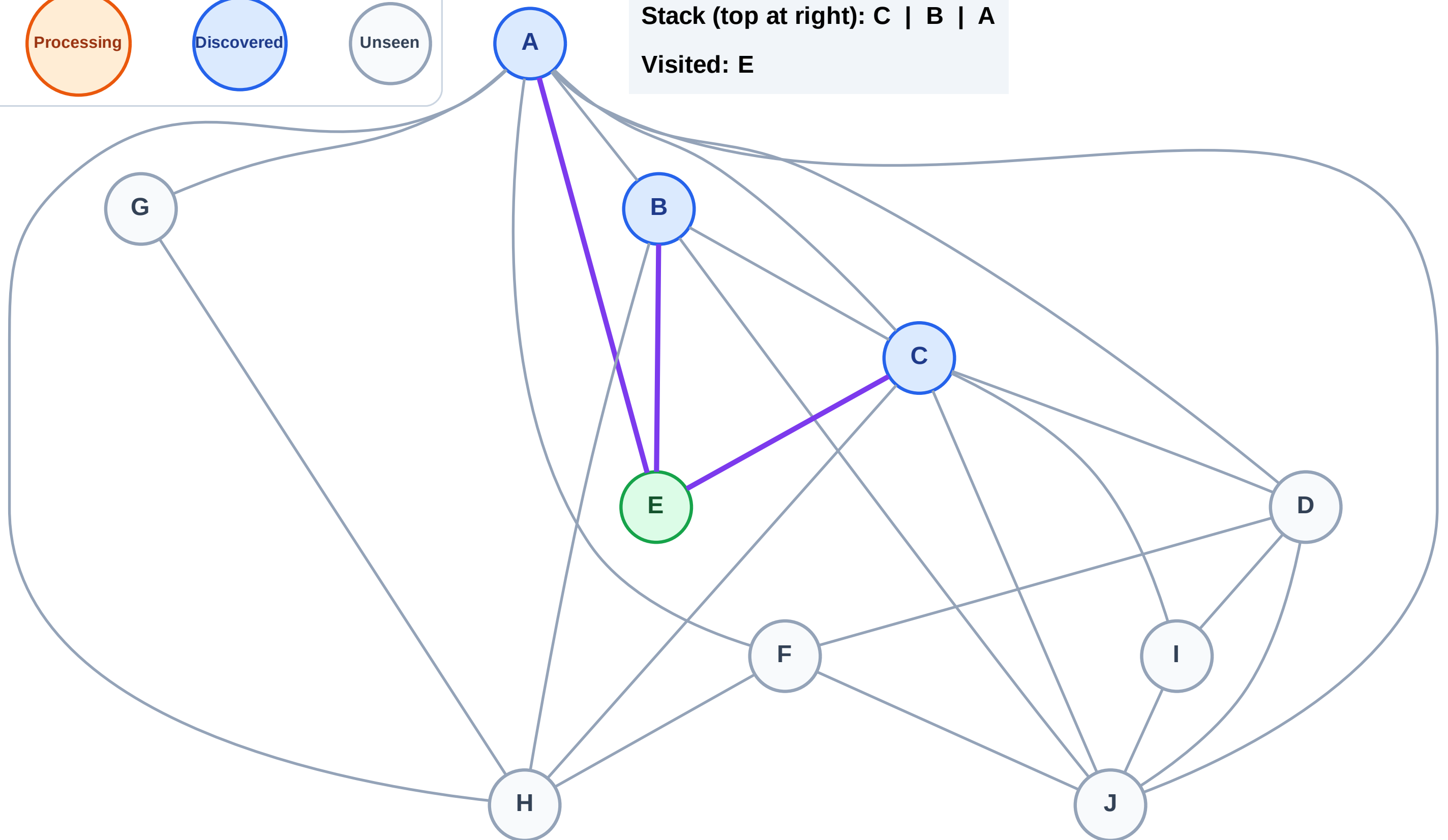
Step 3: Discover C, B, A from E

Legend



Stack (top at right): C | B | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

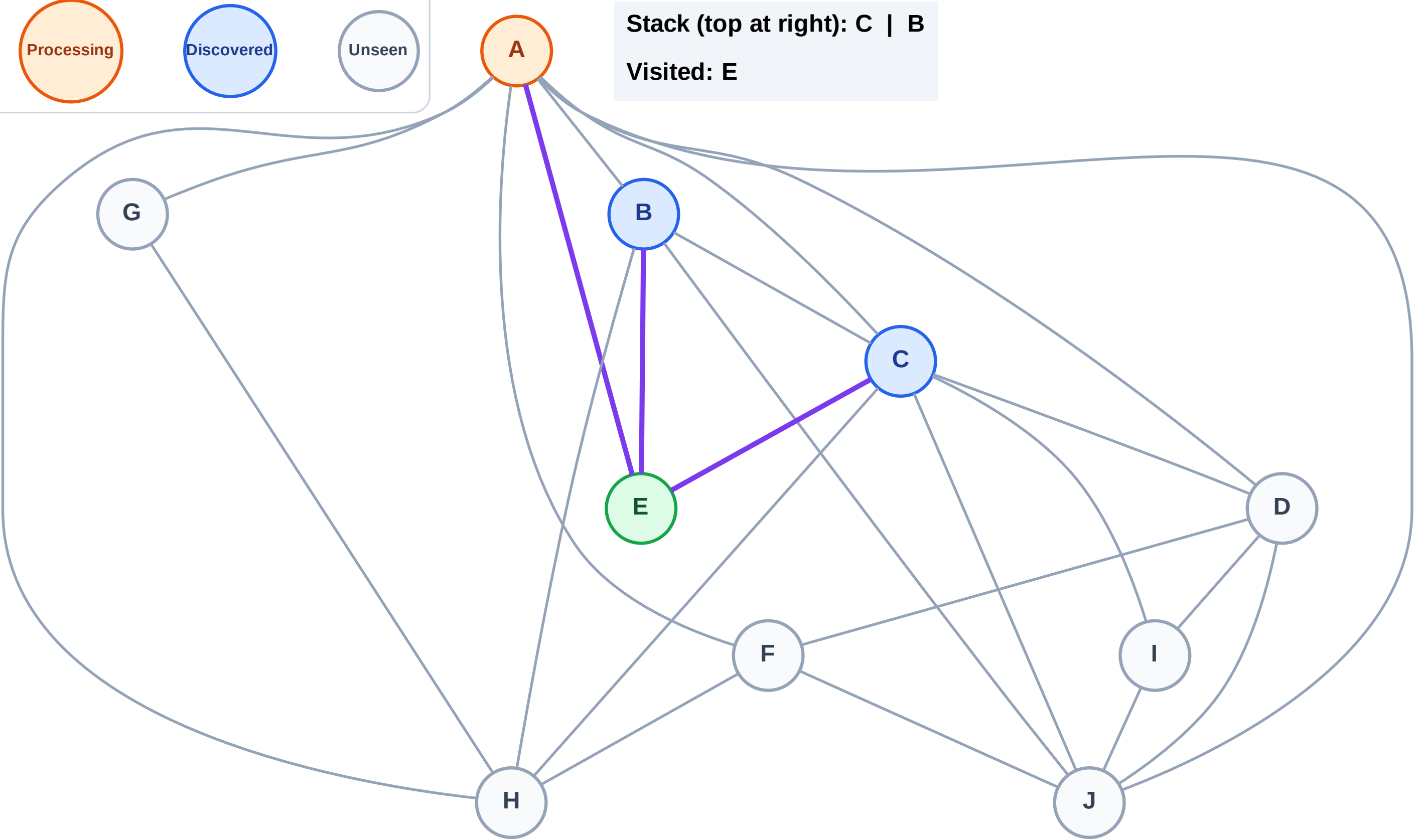
Complete

Processing

Discovered

Unseen

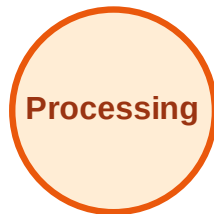
Stack (top at right): C | B
Visited: E



DFS Traversal

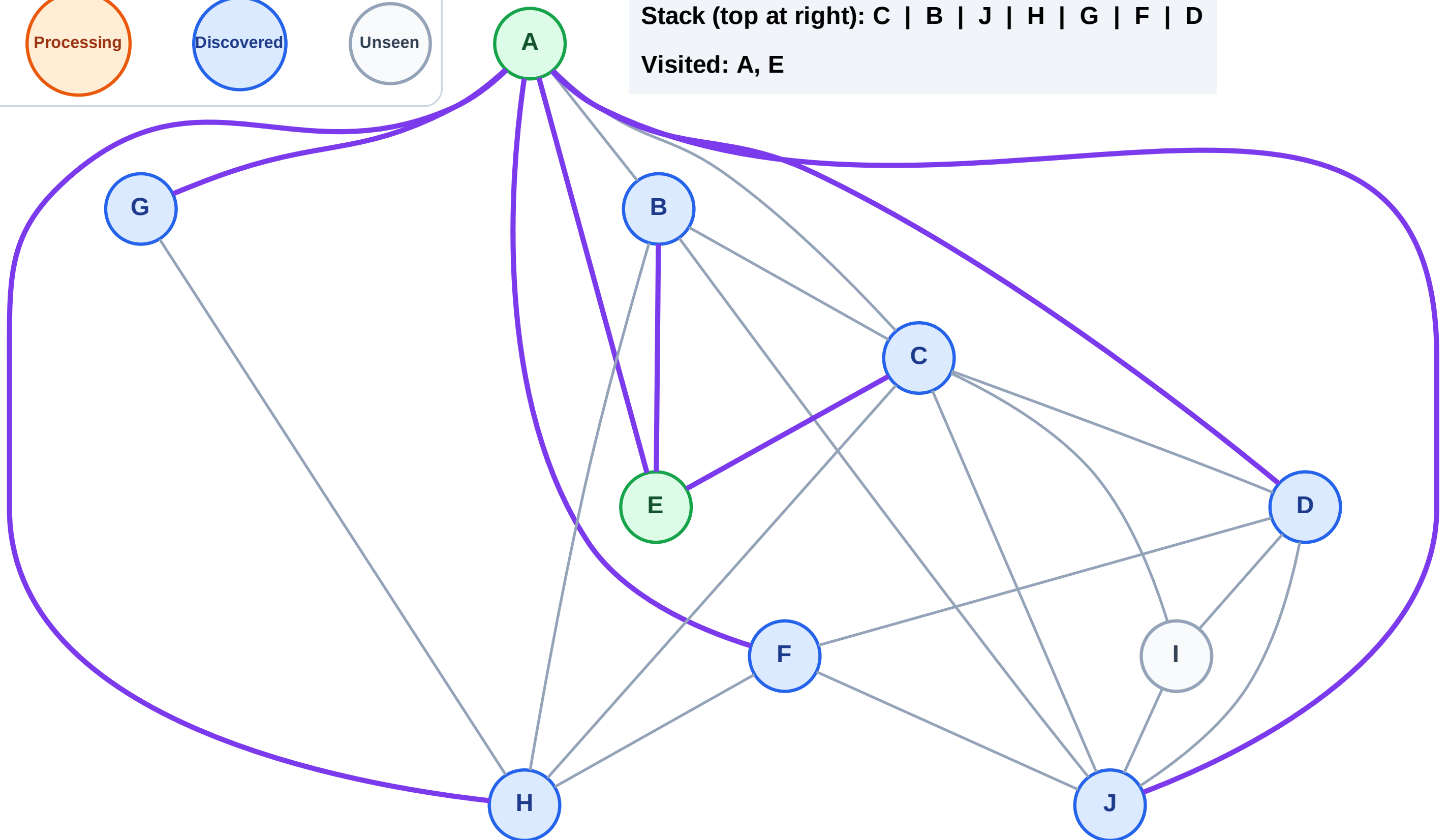
Step 5: Discover J, H, G, F, D from A

Legend



Stack (top at right): C | B | J | H | G | F | D

Visited: A, E



DFS Traversal

Step 6: Process node D

Legend

Complete

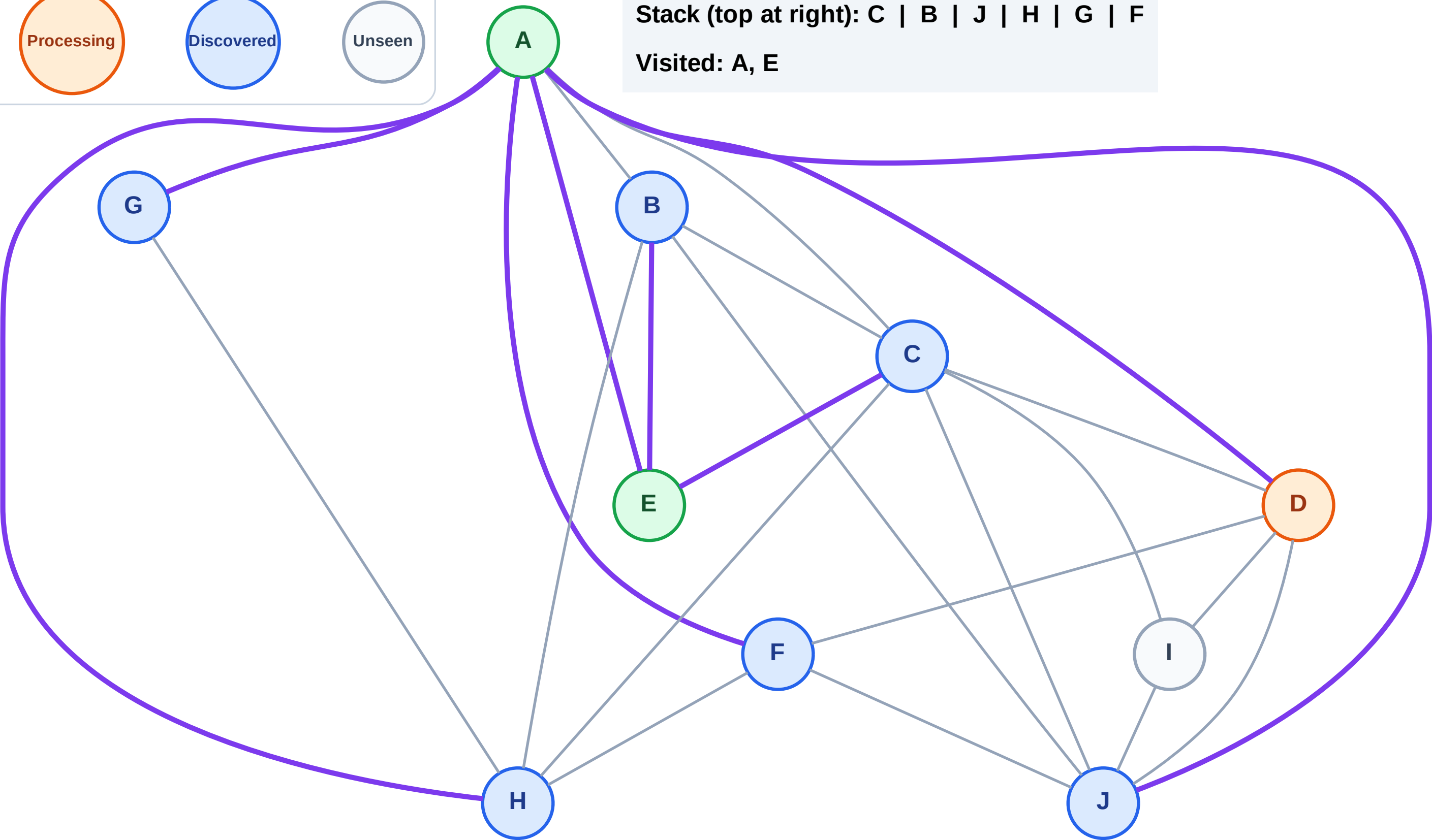
Processing

Discovered

Unseen

Stack (top at right): C | B | J | H | G | F

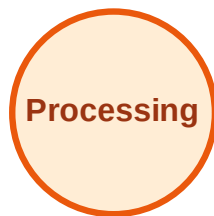
Visited: A, E



DFS Traversal

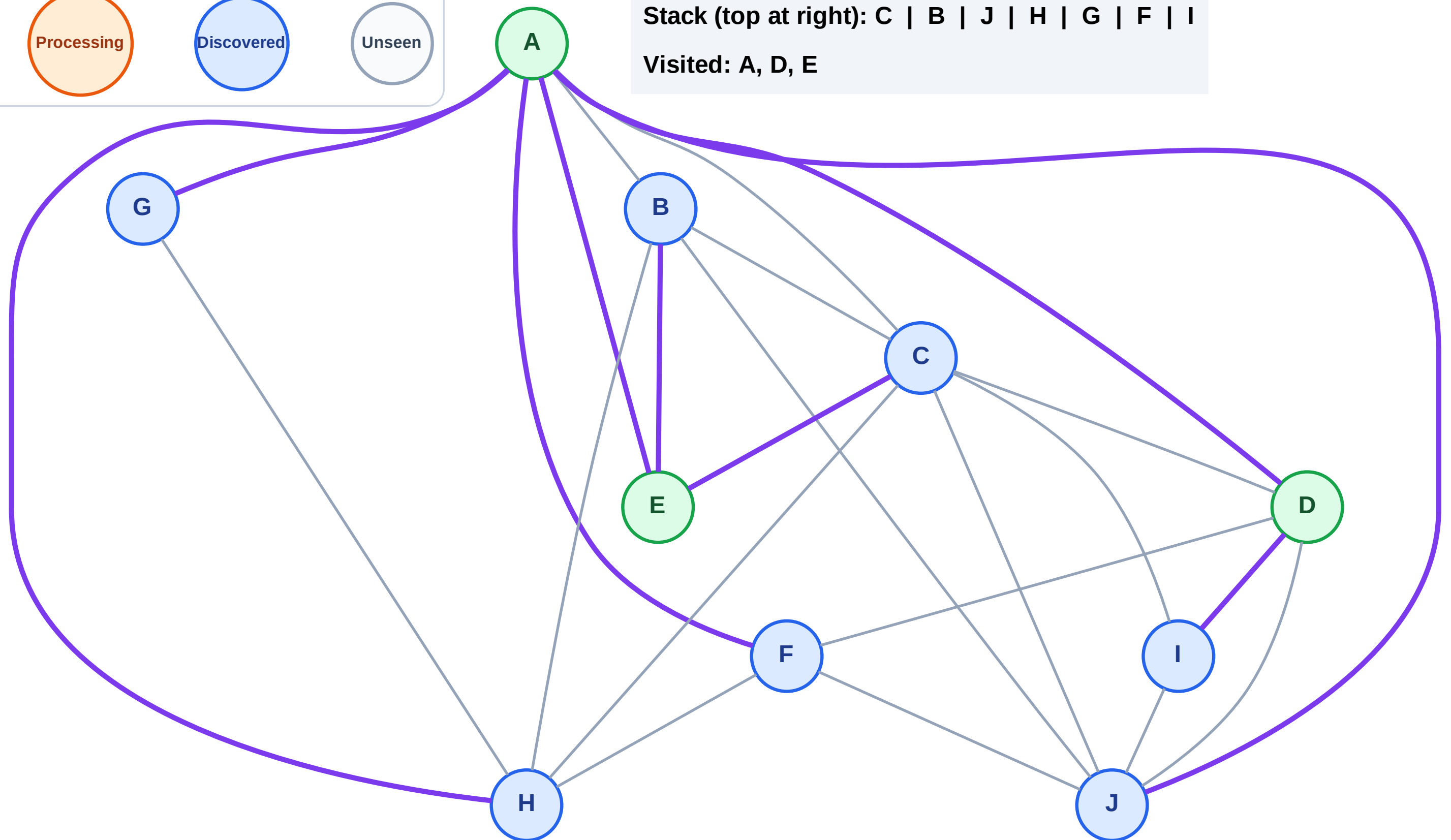
Step 7: Discover I from D

Legend



Stack (top at right): C | B | J | H | G | F | I

Visited: A, D, E



DFS Traversal

Step 8: Process node I

Legend

Complete

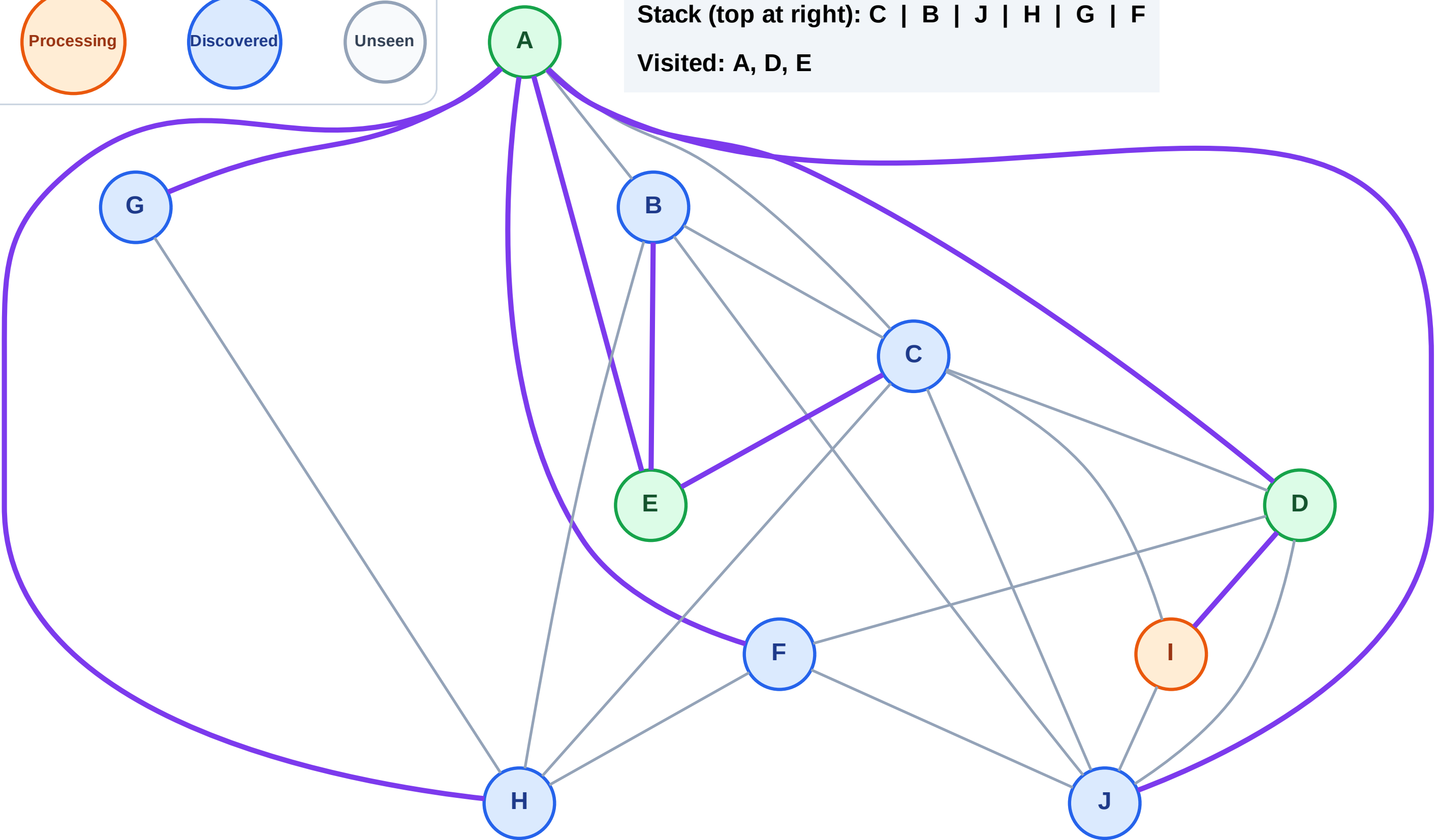
Processing

Discovered

Unseen

Stack (top at right): C | B | J | H | G | F

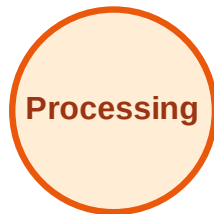
Visited: A, D, E



DFS Traversal

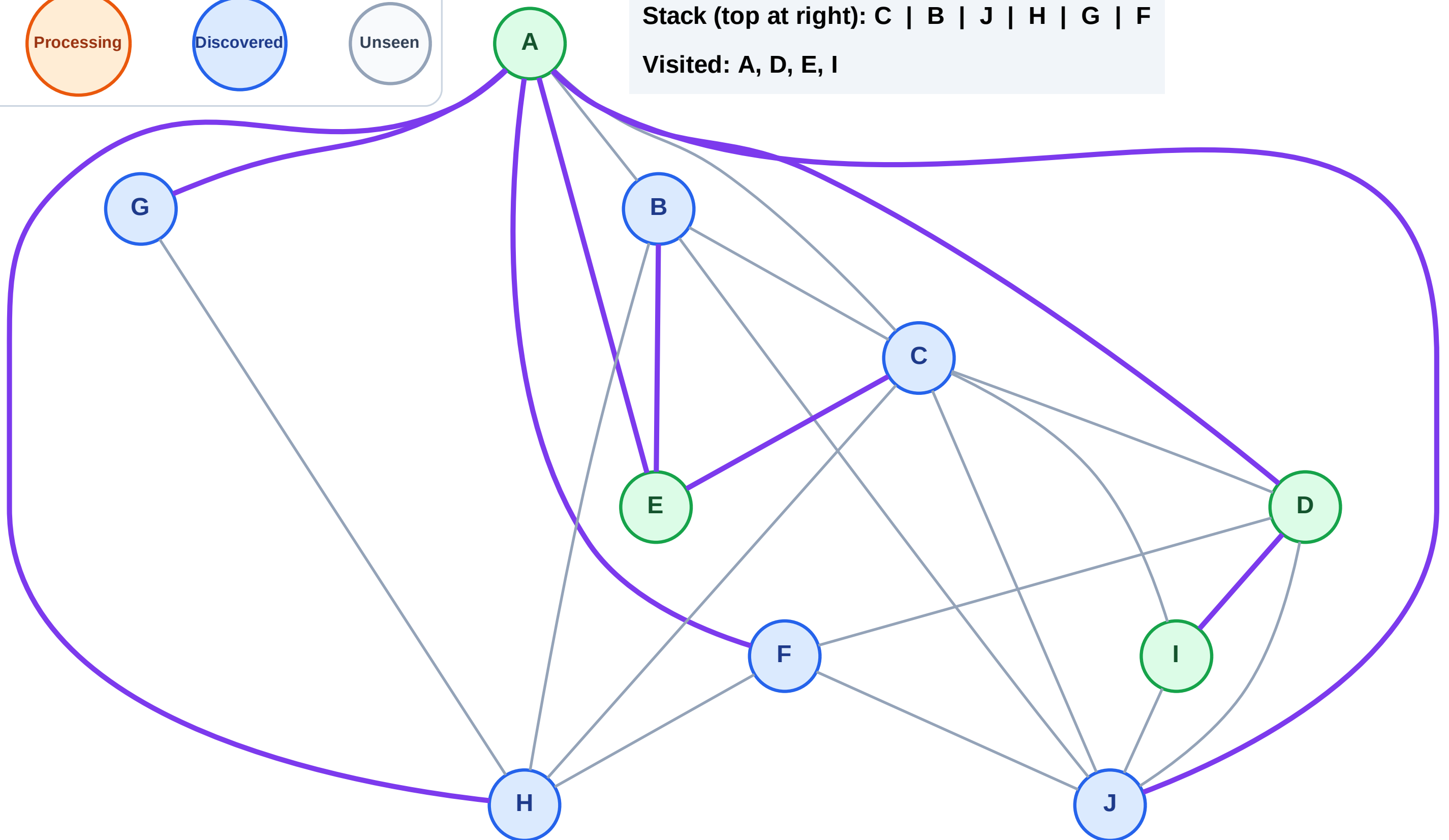
Step 9: No new neighbors from I

Legend



Stack (top at right): C | B | J | H | G | F

Visited: A, D, E, I



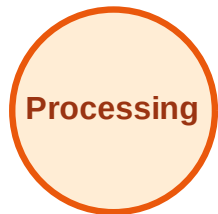
DFS Traversal

Step 10: Process node F

Legend



Complete



Processing



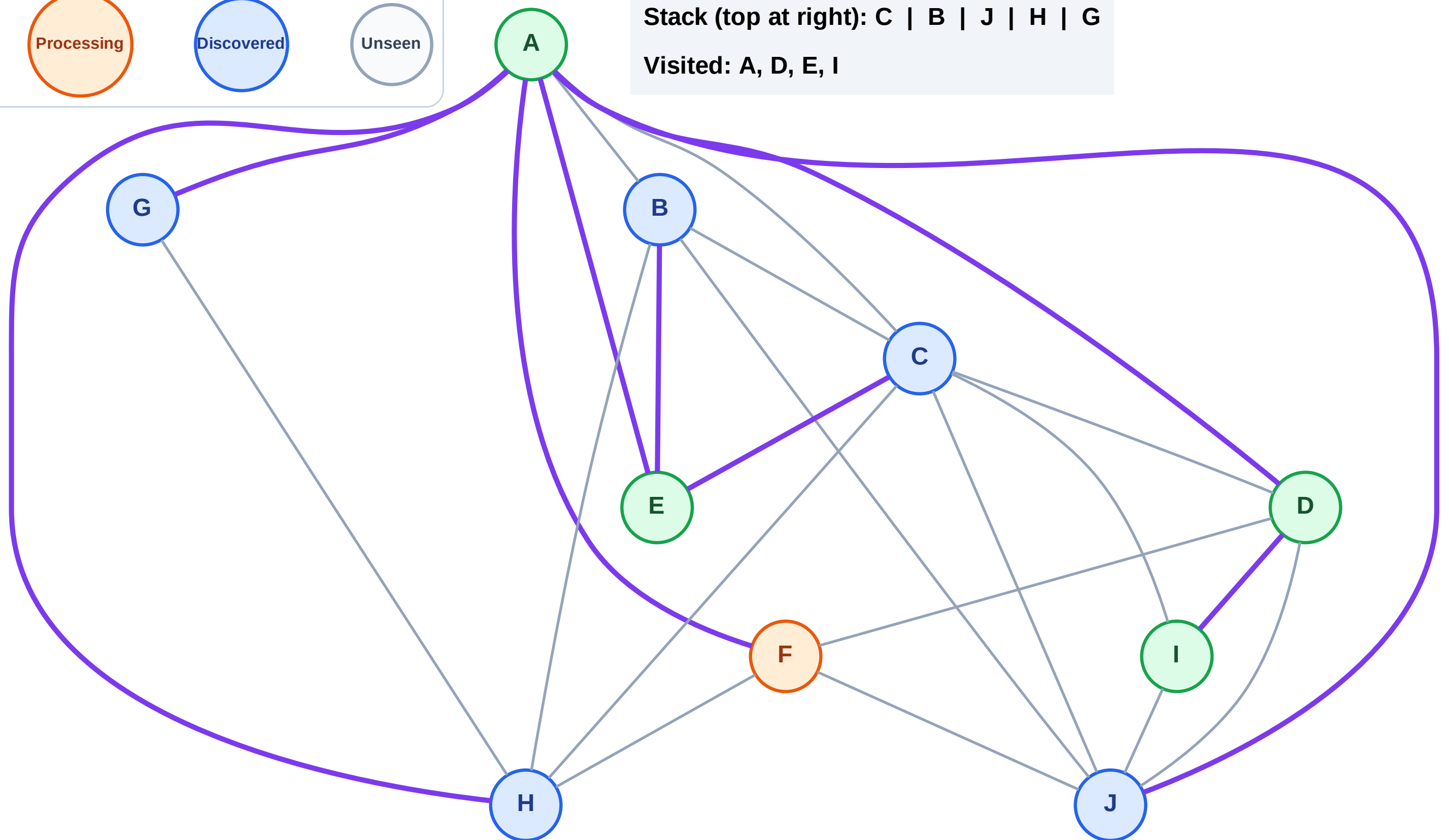
Discovered



Unseen

Stack (top at right): C | B | J | H | G

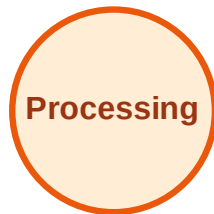
Visited: A, D, E, I



DFS Traversal

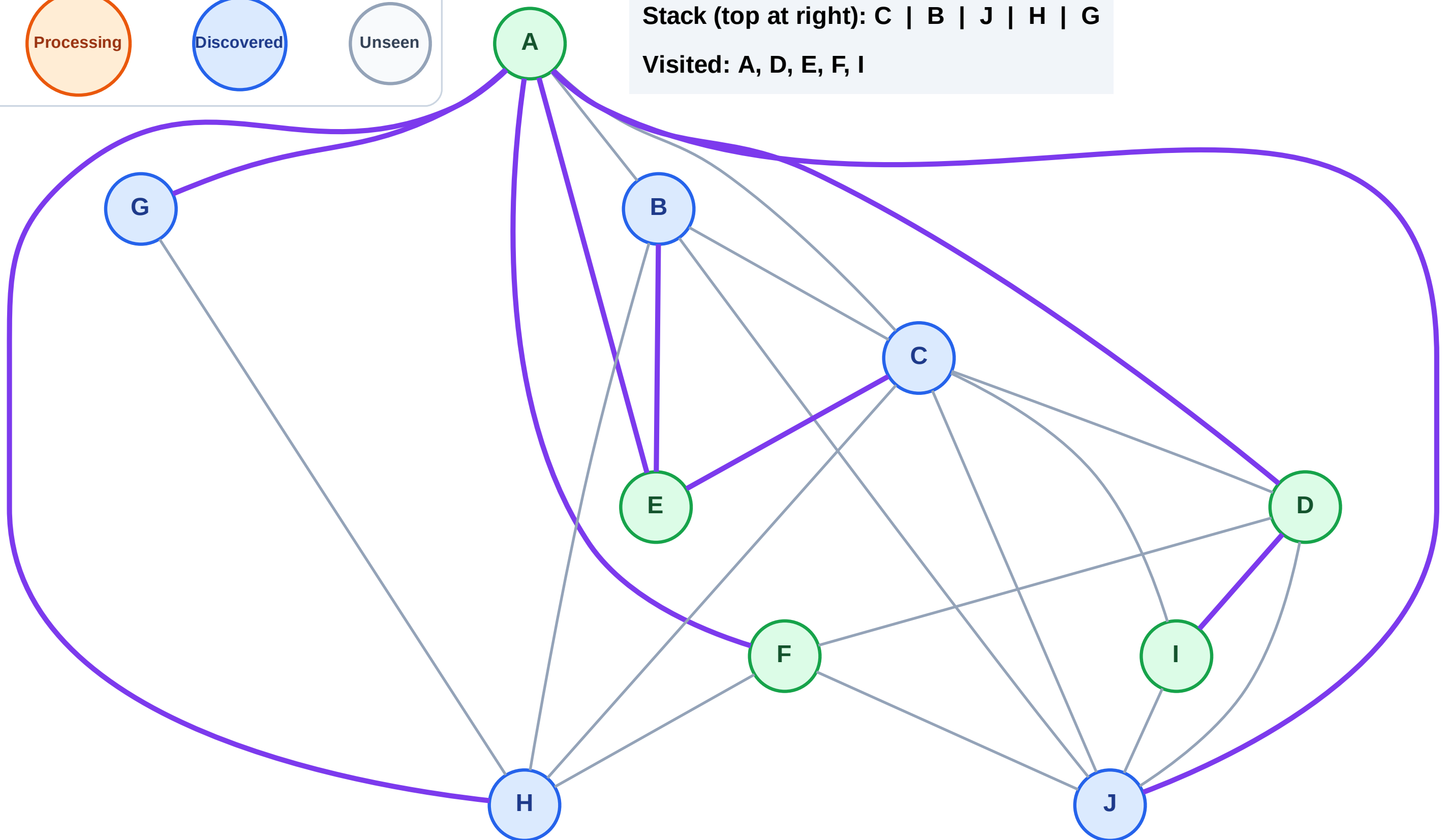
Step 11: No new neighbors from F

Legend



Stack (top at right): C | B | J | H | G

Visited: A, D, E, F, I



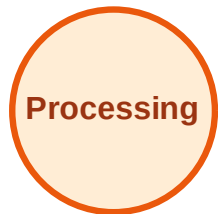
DFS Traversal

Step 12: Process node G

Legend



Complete



Processing



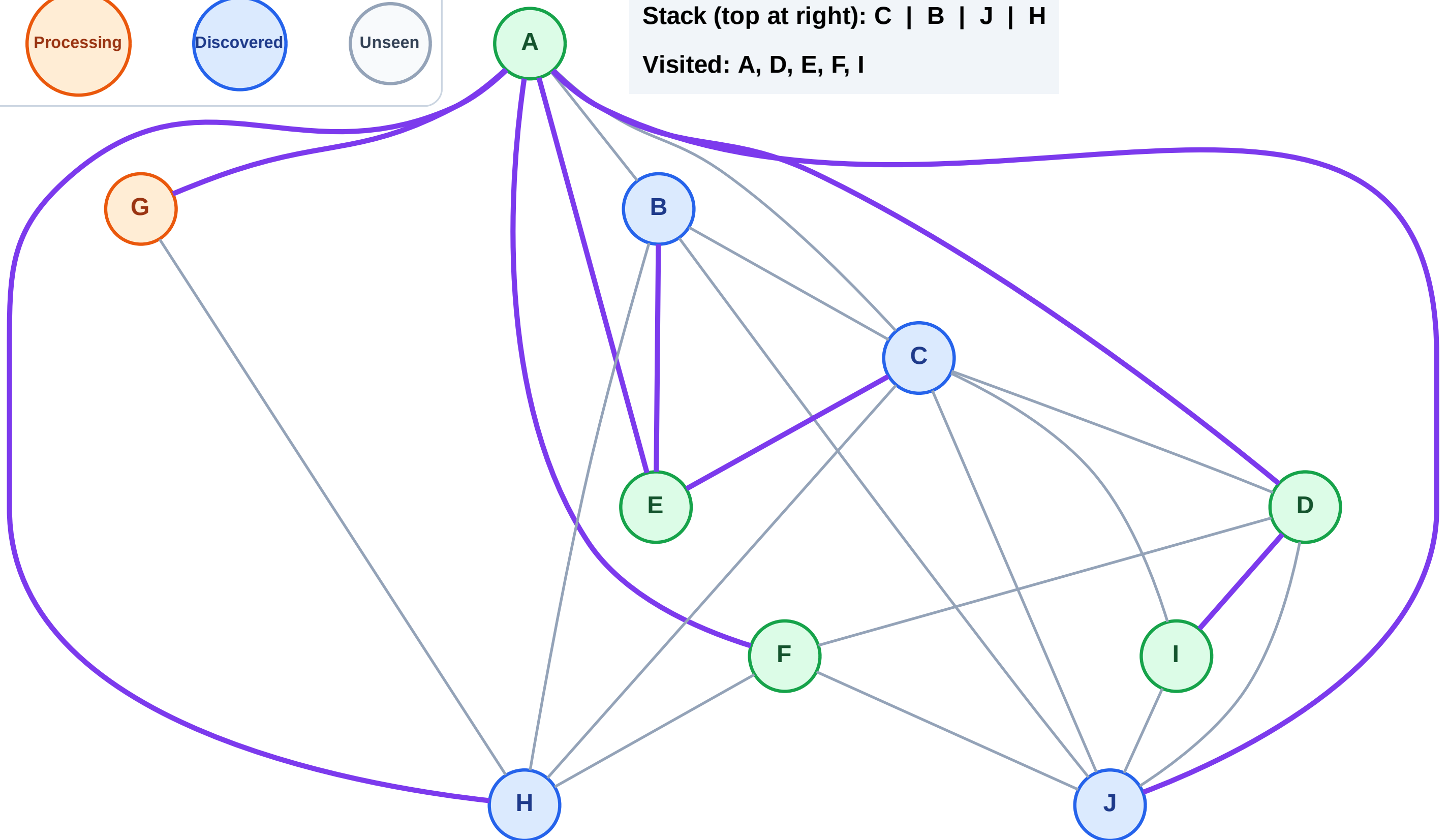
Discovered



Unseen

Stack (top at right): C | B | J | H

Visited: A, D, E, F, I



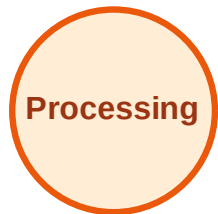
DFS Traversal

Step 13: No new neighbors from G

Legend



Complete



Processing



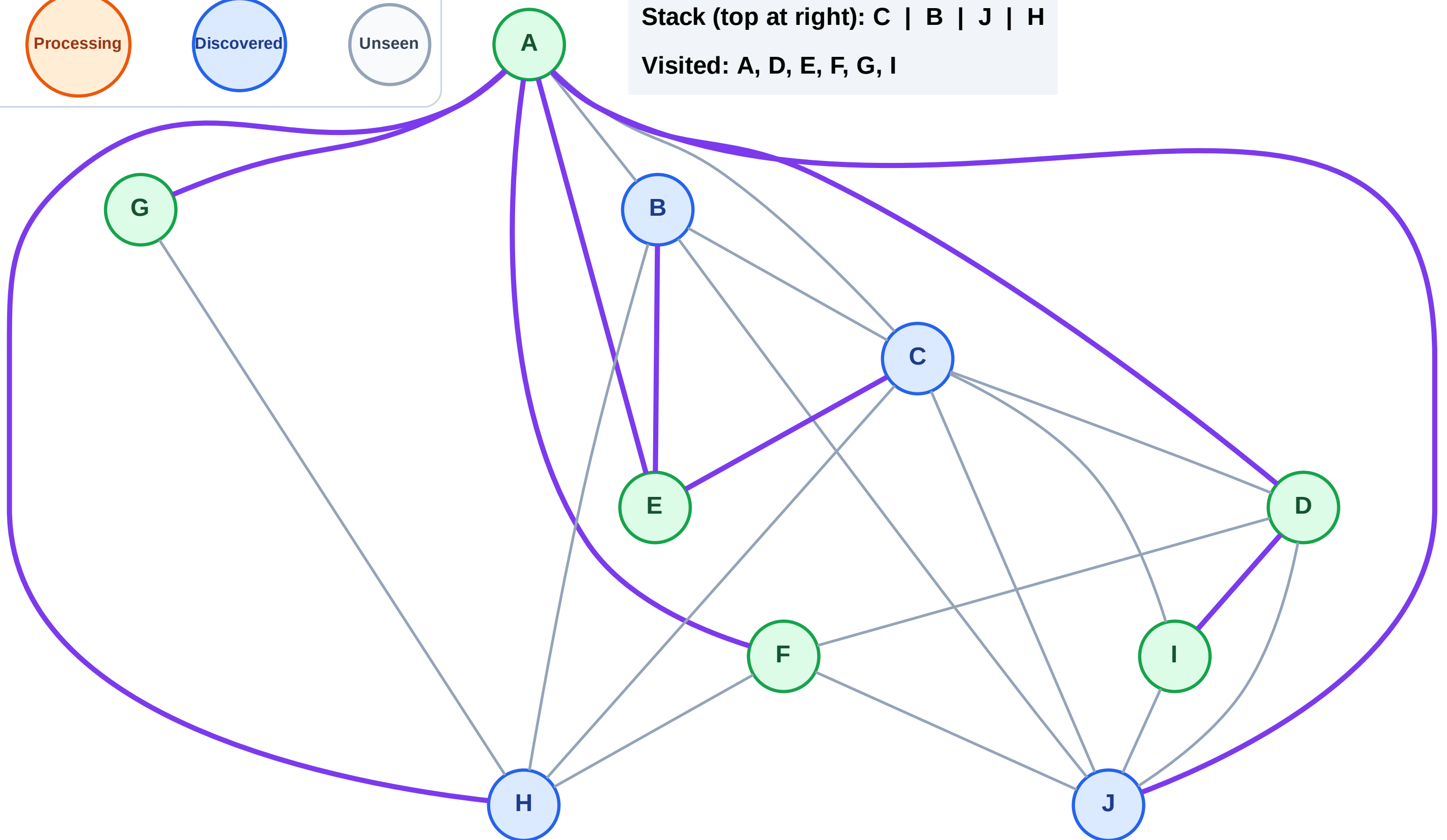
Discovered



Unseen

Stack (top at right): C | B | J | H

Visited: A, D, E, F, G, I



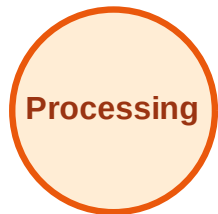
DFS Traversal

Step 14: Process node H

Legend



Complete



Processing



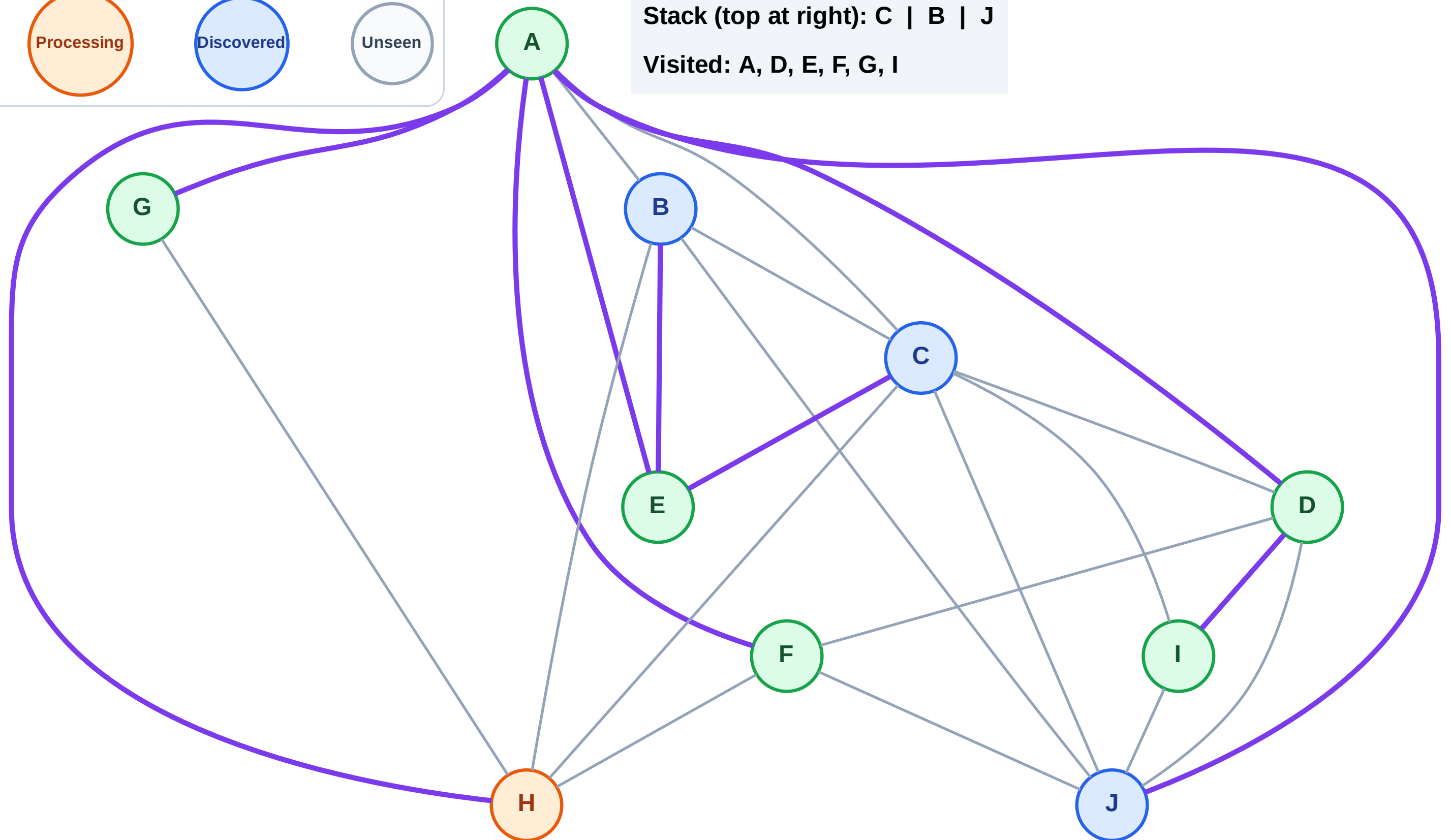
Discovered



Unseen

Stack (top at right): C | B | J

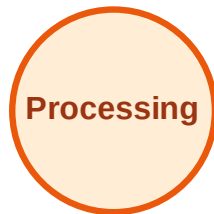
Visited: A, D, E, F, G, I



DFS Traversal

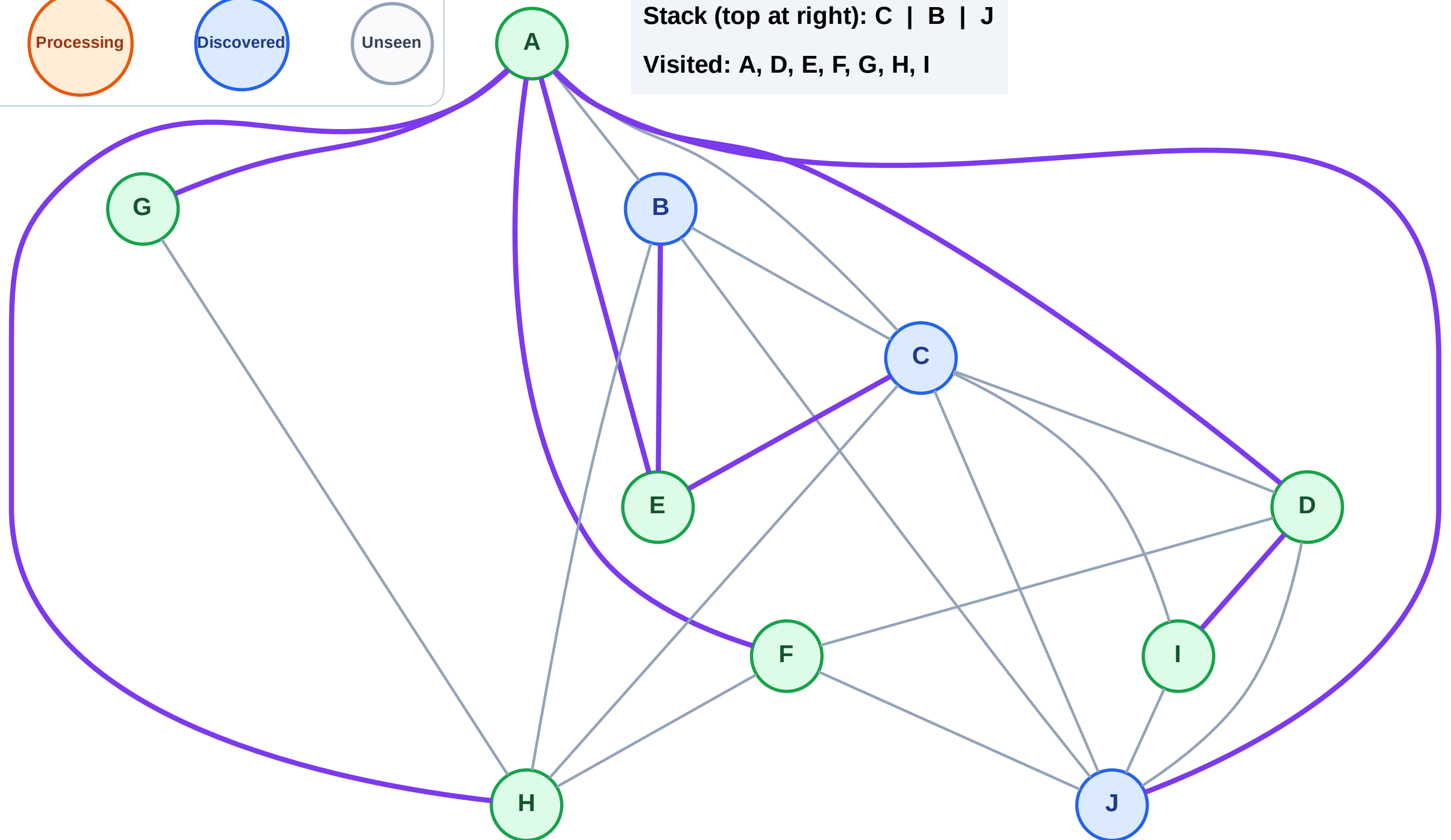
Step 15: No new neighbors from H

Legend



Stack (top at right): C | B | J

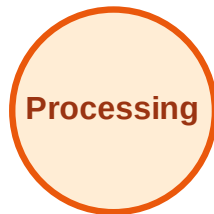
Visited: A, D, E, F, G, H, I



DFS Traversal

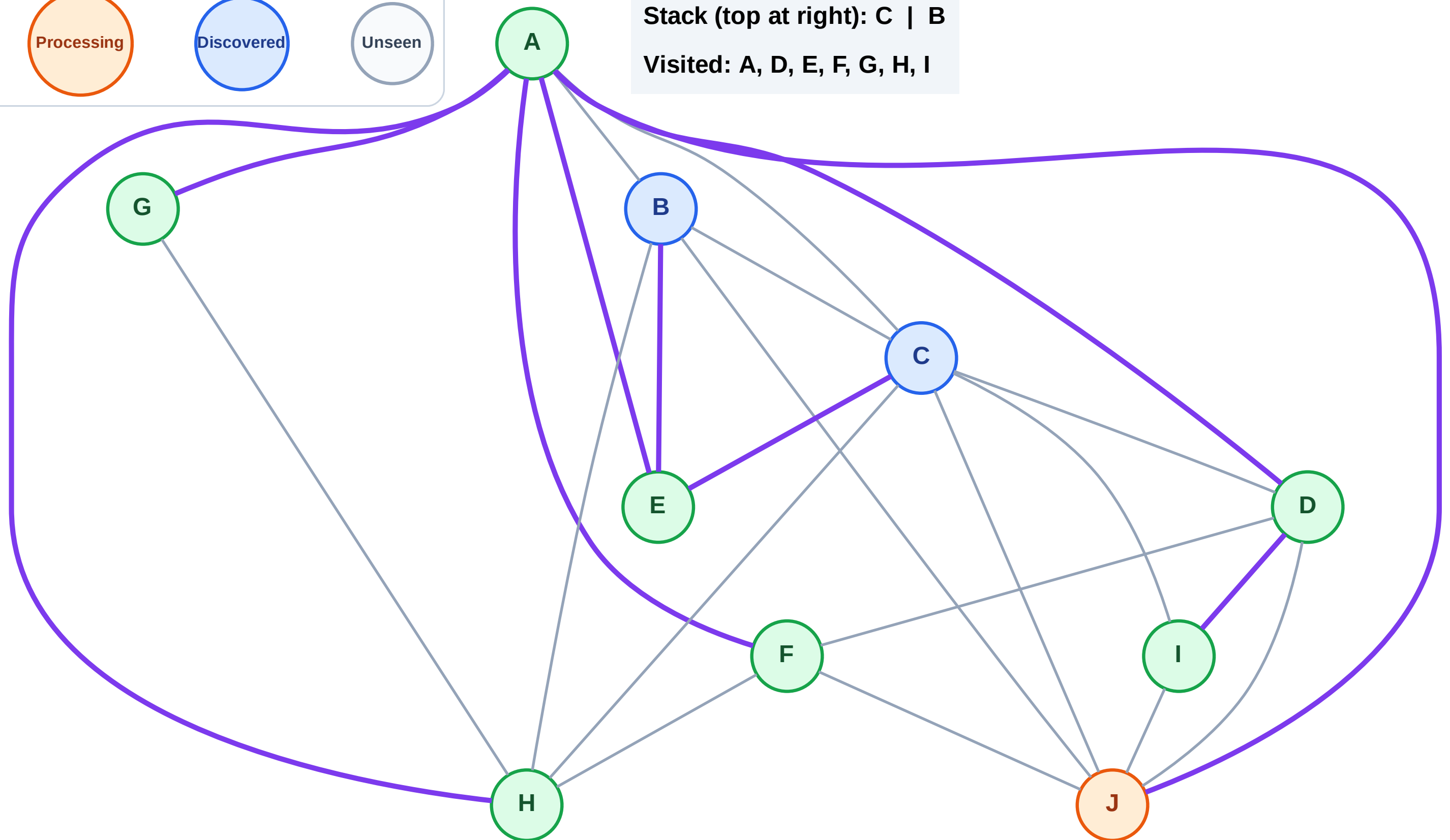
Step 16: Process node J

Legend



Stack (top at right): C | B

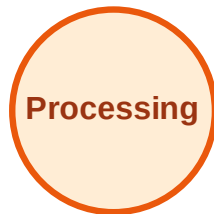
Visited: A, D, E, F, G, H, I



DFS Traversal

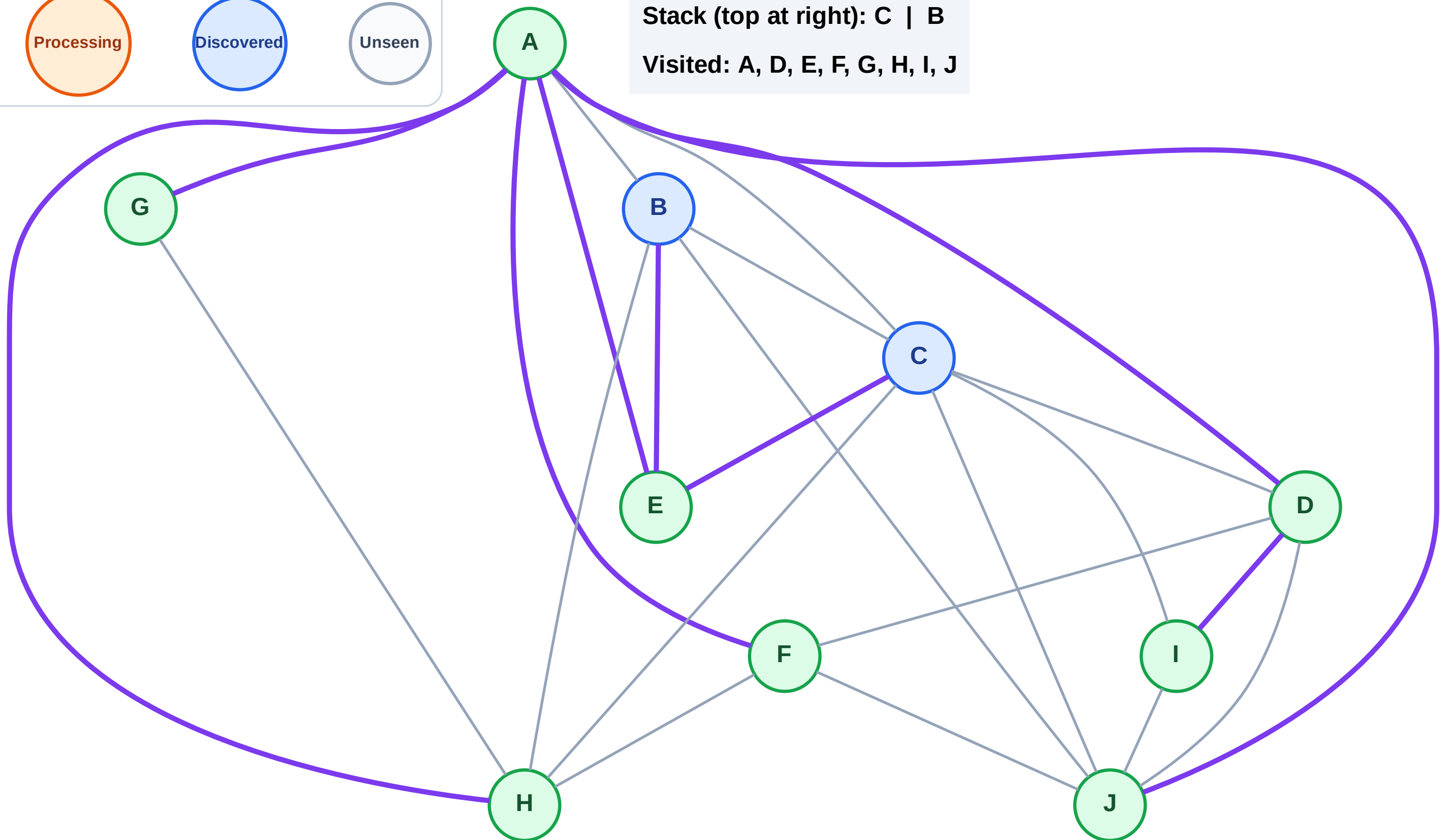
Step 17: No new neighbors from J

Legend



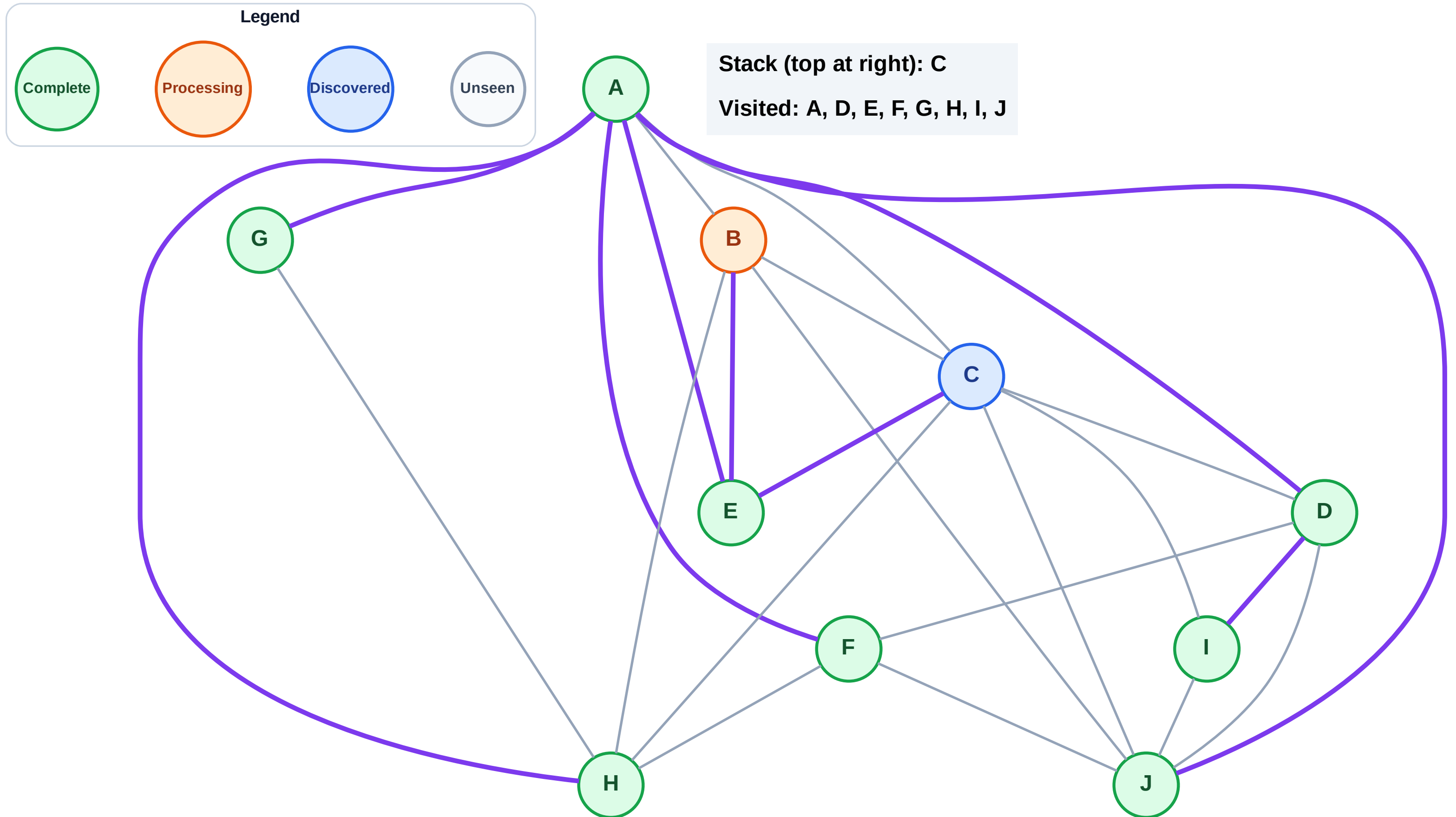
Stack (top at right): C | B

Visited: A, D, E, F, G, H, I, J



Step 18: Process node B

Step 18: Process node B



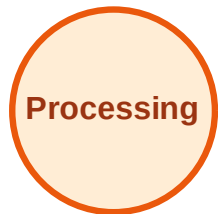
DFS Traversal

Step 19: No new neighbors from B

Legend



Complete



Processing



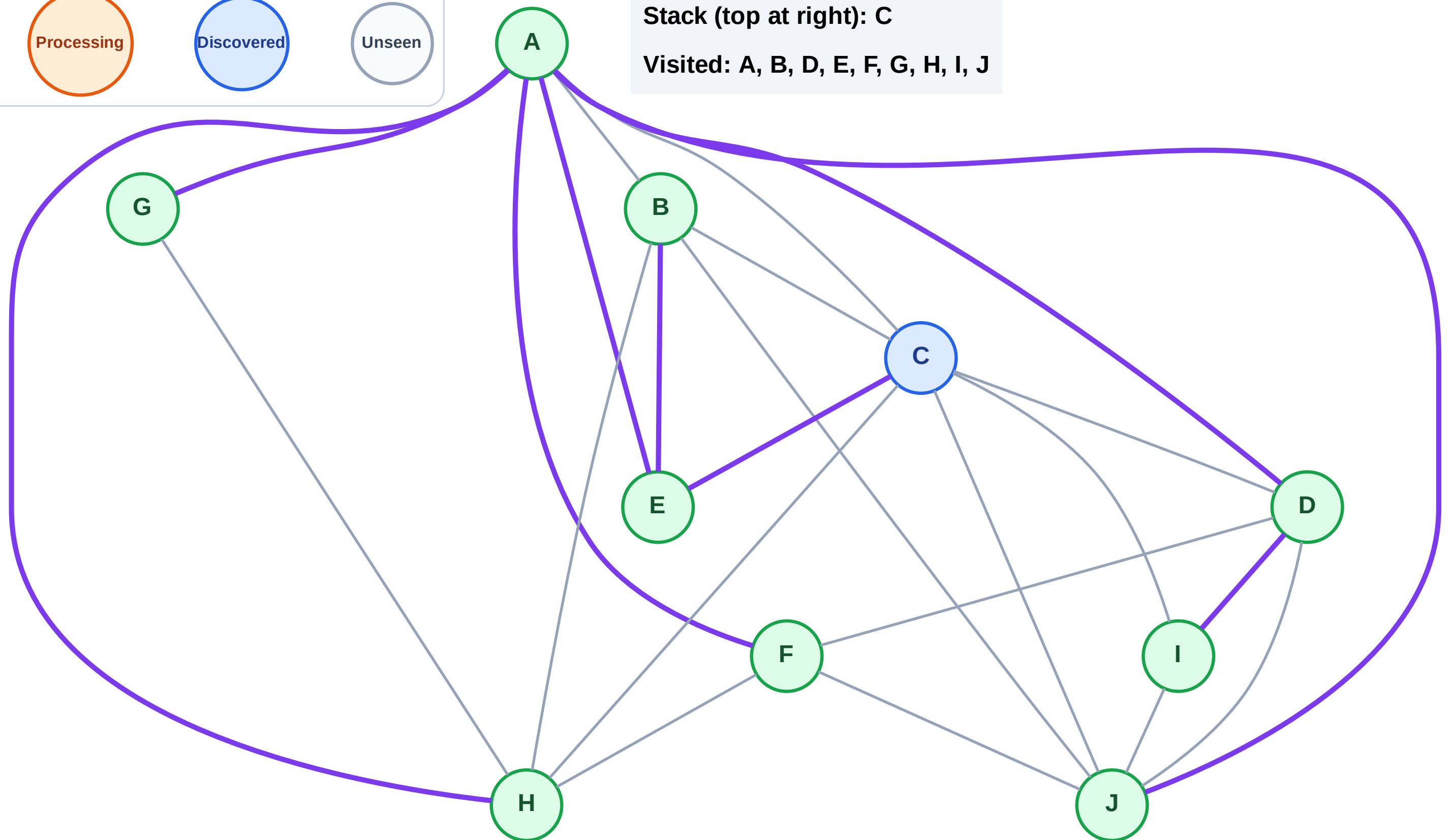
Discovered



Unseen

Stack (top at right): C

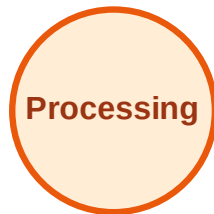
Visited: A, B, D, E, F, G, H, I, J



DFS Traversal

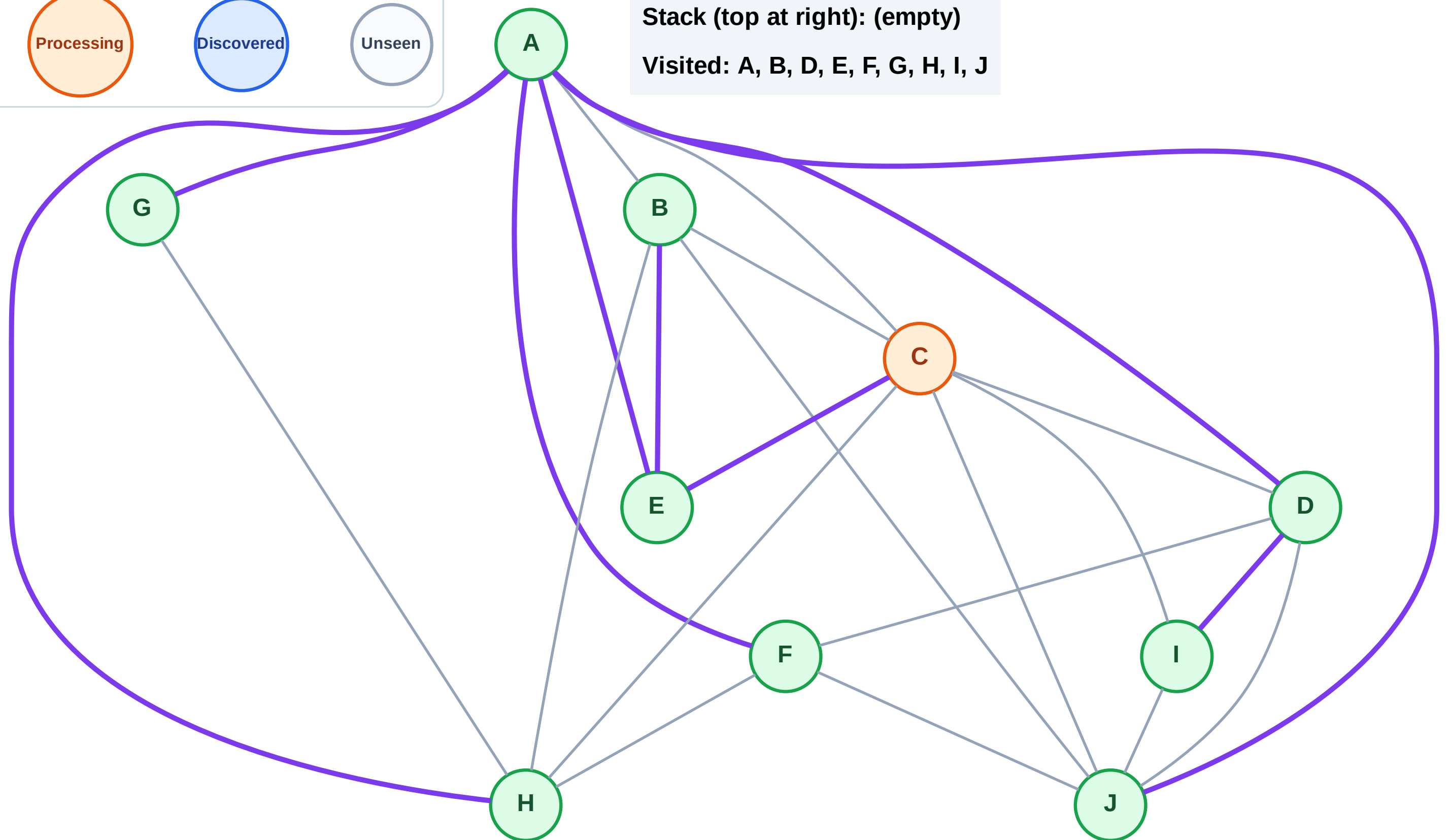
Step 20: Process node C

Legend



Stack (top at right): (empty)

Visited: A, B, D, E, F, G, H, I, J



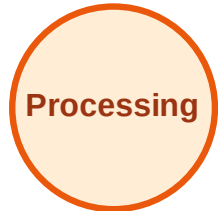
DFS Traversal

Step 21: No new neighbors from C

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

